Two barriers for minimal-time firing squads, with applications to four and five states

Kia Keng Giam
Temasek Junior College, Singapore
giamkiakeng@gmail.com

Abstract

A minimal-time solution of the firing squad synchronization problem synchronizes every line of n cells at time $2n-2$. Six states suffice; whether five do has been open since 1987. Every line agrees with the half-line until its right end's echo returns. For any number of states, we prove that cells at distance j behind the half-line's front become periodic in time only after time $(3/2-\varepsilon)j$, for every $\varepsilon > 0$ and infinitely many j, and that the half-line is never periodic in time and space on a wedge at its left border, hence not eventually regular in our sense. With certified satisfiability checks, we exclude four states using the lines of lengths 2 to 5 and parts of the next five, and show that every five-state solution uses at least 15 non-quiescent neighbourhoods on the first 78 anti-diagonals of its half-line. We give five-state rules synchronizing exactly the lengths 2 to 14.

Keywords: firing squad synchronization problem; cellular automata; minimal time; state complexity; lower bounds; pumping lemma; eventual periodicity; SAT solving; proof certificates

1 INTRODUCTION

The firing squad synchronization problem, posed by Myhill in 1957 and made popular by Moore [1], asks for a finite automaton with the following property. Copies of the automaton are placed in the cells of a line of arbitrary length n. At time 0 the leftmost cell, the *general*, is in a special state and all other cells are *quiescent*. At each step every cell updates its state from its own state and the states of its two neighbours. All cells must then enter a *firing* state at the same time, and none of them earlier. The difficulty is that the cells are identical, that each sees only its two neighbours, and that a cell with k states cannot count up to n: the same finite rule has to synchronize lines of every length.

No line can be synchronized before time $2n-2$: the order of the general needs $n-1$ steps to reach the far end of the line, and the news that the far end exists needs $n-1$ more steps to come back. Goto [2] and Waksman [3] showed that time $2n-2$ can be achieved, and a solution that synchronizes every line of length $n \geq 2$ at time $2n-2$ is called a *minimal-time solution*. Since then, the least number of states of a minimal-time solution has been a benchmark question of cellular automata theory [4, 5, 6]. Waksman's solution has 16 states, Balzer's 8 [7], Gerken's 7 [8] and Mazoyer's 6 [9]. In the other direction, Balzer [7] reported a computer search showing that no four-state minimal-time solution exists; Yunès [10] repeated the search and observed that the lines of lengths $2, \ldots, 9$ suffice, and Sanders [11] found Balzer's search incomplete and confirmed the result with a corrected one. Many six-state minimal-time solutions are known today [12, 13]; solutions with at most five states are known for restricted families of lengths, such as the powers of two, some of them in non-minimal time [14, 15], and four-state protocols synchronize the rings whose lengths are powers of two [16]. Whether a five-state minimal-time solution exists has been open since Mazoyer's construction [5, 6].

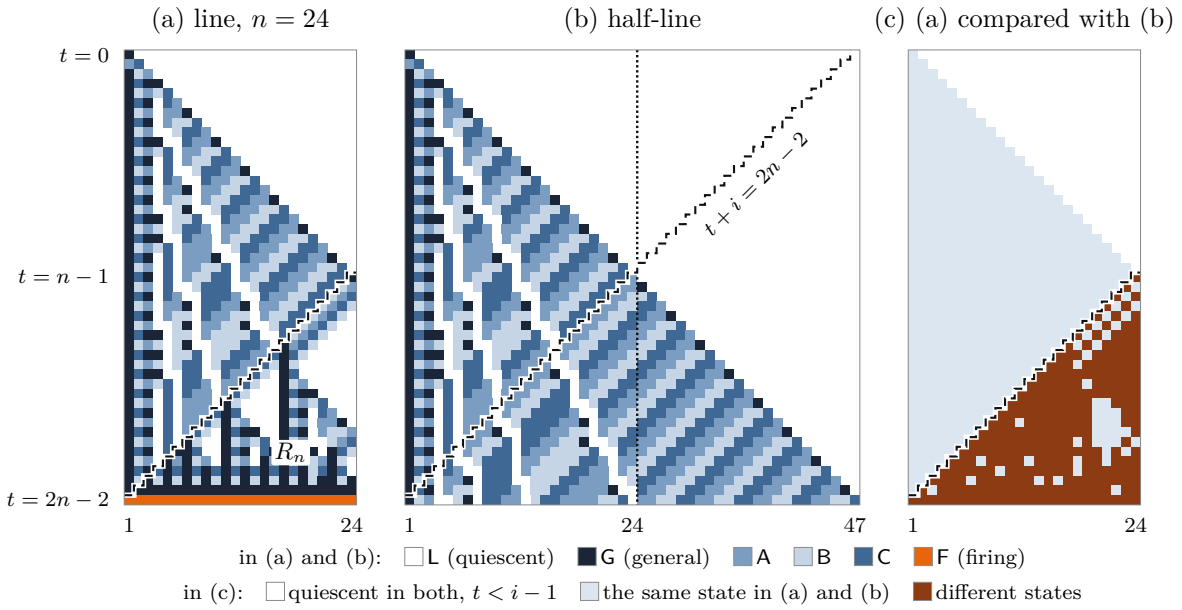


Figure 1: Mazoyer's six-state minimal-time solution. (a) The line of length $n = 24$; time runs downwards from $t = 0$ to the firing time $t = 2n - 2 = 46$, and the cells $i = 1, \ldots, 24$ run from left to right. (b) The half-line, which is started in the same way but has no right end, for the same times; the dotted line marks the right end of the line in (a). (c) Each cell of (a) compared with the same cell of (b). Above the front $t = i - 1$, both are quiescent (Lemma 2.1). The dashed staircase separates the cells with $t + i \leq 2n - 2$, on which (a) and (b) coincide (Lemma 2.2), from the reflected triangle R_n, in which the line reacts to its right end. On the first anti-diagonal of R_n, the *return chain* $t + i = 2n - 1$, every cell of the line differs from the same cell of the half-line (Lemma 2.4); further down, the two diagrams coincide on some cells and differ on others.

Why lower bounds are hard A five-state rule is a table of 96 entries (Section 2), two of which are fixed by quiescence, so there are $5^{94} > 10^{65}$ rules. A rule can be refuted by a single line that it fails to synchronize, but a minimal-time solution has to synchronize every line, so a proof that five states do not suffice must exclude all rules. As there are finitely many rules, it suffices to find one length N such that no five-state rule synchronizes the lines of lengths $2, \ldots, N$; a negative answer can thus consist of finitely many conditions. The difficulty is to find conditions that are strong enough and cheap enough to check. This paper proves such conditions for every number of states and measures how far they reach for four and five states.

The idea Write (t, i) for the cell i at time t. The *half-line* is started like a line, with the general in cell 1, but has no right end. Until the echo of its right end comes back, a line cannot tell that it is finite: the line of length n and the half-line agree on every cell (t, i) with $t + i \leq 2n - 2$ (Lemma 2.2; Figure 1 shows this for Mazoyer's solution). The line can therefore differ from the half-line only on the cells (t, i) with $t + i \geq 2n - 1$ and $t \leq 2n - 2$, which form a triangle next to the right end, the *reflected triangle*. So a minimal-time solution consists of one space-time diagram of the half-line, shared by all lengths, and, for each length n, a reflected triangle in which the line reacts to its right end. Everything that the reflected triangle of the line of length n learns from the half-line is contained in the cells of the half-line with $t + i = 2n - 3$ or $t + i = 2n - 2$ (Lemma 3.1). Conditions on the half-line alone are therefore cheap to check, and they apply to all lengths at once.

Results Our main results concern minimal-time solutions with any number of states.

(1) *A late onset of periodicity* (Theorem 3.4). The front of the half-line is the cell $(t, t+1)$ reached by the general's signal at time t. For $j \geq 0$, the cells $(t, t+1-j)$ with $t \geq j$, which stay at distance j behind the front, form the *depth row* j. In a minimal-time solution every depth row is eventually periodic in time; let $\theta(j)$ be the time from which depth row j is periodic. We prove that $\limsup_{j\to\infty} \theta(j)/j \geq 3/2$: for every $\varepsilon > 0$ there are infinitely many j with $\theta(j) \geq (3/2 - \varepsilon)j$. The proof rests on a pumping lemma (Lemma 3.3): if the reflected triangles of two lines of different lengths received the same values on the cells that determine their right ends, these ends would fire at the same number of steps after the front reaches them, which is impossible (Figure 5). At the time $\theta(j)$, depth row j is at the cell $\theta(j) + 1 - j$, which is close to the cell $\theta(j)/3$ or beyond when $\theta(j) \geq (3/2 - \varepsilon)j$; for this reason we also call the theorem the *speed*-1/3 *barrier*. The name describes this bound only: the theorem concerns infinitely many depth rows, not all of them, and it does not assert that a boundary or a signal moves at speed 1/3. In the known constructions, the depth rows become periodic where they cross a signal of the general, and if that is so, the bound says that the first division point of the line is asymptotically not in the half next to the general; this reading is a heuristic (Remark 3.5). In simulations up to time 1400, the depth rows of the five-state rule δ_{12} of Section 6.3, which synchronizes the lines of lengths $2, \ldots, 12$, become periodic at the time $\lceil 3j/2 \rceil + 2$, at the bound (Figure 6).

(2) *The left-border barrier* (Theorem 4.2 and Corollary 4.3). The half-line is never periodic in time and space on a wedge $\{(t, i) : s \leq i \leq \alpha t,\ t \geq t_0\}$ at its left border. In particular it is not *eventually regular* in the sense of Section 4: from some time on, it does not consist of finitely many periodic blocks, separated by signals of rational speeds, whose lengths grow linearly. Other kinds of regular behaviour, and periodic regions away from the left border, are not excluded. The reason is that the cells near the left border are fired by a finite automaton that reads the half-line along the cells with $t + i = 2n - 3$ and $t + i = 2n - 2$, and a long periodic input would make it fire too early (Lemma 4.1 and Figure 10). This excludes the half-line of δ_{12} for the line of length 518 (Corollary 6.5 and Figure 17).

(3) *Finitely many early rows* (Corollary 5.3). The front reaches the right end of the line of length n at time $n - 1$. For a fixed T, the cells of the reflected triangles at the times $n - 1, \ldots, n - 1 + T$, which we call the first $T + 1$ rows, take only finitely many forms over all lengths n, and a certificate computed from the half-line determines them. This turns the condition that no line fires during these $T + 1$ steps into finitely many checks (Lemma 5.5 and Figure 12).

With these conditions and satisfiability solvers whose negative answers are certified by independently checked proofs, we obtain the following results. Four states are excluded by the lines of lengths $2, \ldots, 5$ together with the first five rows, the return chains and the right ends of the lines of lengths $6, \ldots, 10$, and this is sharp in several directions (Theorem 6.1 and Proposition 6.2). Five five-state rules synchronize exactly the lines of lengths $2, \ldots, 14$ (Proposition 6.3), so the lines of lengths $2, \ldots, 14$ alone cannot exclude five states; for four states, the longest such range is $2, \ldots, 8$ (Proposition 6.2). On the cells of its half-line with $t + i \leq 78$, every five-state minimal-time solution uses at least 15 distinct neighbourhoods besides the quiescent one (Proposition 6.4). Finally, a conditional five-state result reported by Balzer in 1967 holds, even in a weaker form (Proposition 6.6). Table 1 lists the results with the kind of evidence behind each of them, and separately the observations from simulations that the paper reports but does not claim beyond the simulated ranges.

Related work Indistinguishability arguments, in which a solution cannot tell two situations apart and therefore fires too early, show that there is no minimal-time solution with one-way communication [17, Prop. 2]. Through repetitions forced by the pigeonhole principle, they

Table 1: The results of this paper, the evidence behind them, and the figures and tables that illustrate them. "Proof" is a proof in the text. A "certified computation" is the unsatisfiability of a propositional formula, with a proof in the LRAT format that an independent checker verifies. An "exhaustive enumeration" or an "exact finite simulation" decides a finite statement completely, and the proof in the text extends it where needed. The last row lists observations from finite simulations, which the paper does not claim beyond the simulated ranges.

Result	Where	Evidence	Shown in
Light cone; agreement with the half-line; return chain	2.1, 2.2, 2.4	proof	Figs. 1–3
Determinacy; cone of the right end	3.1, 3.2	proof	Figs. 4, 5
Pumping lemma; late onset of periodicity (speed-1/3 barrier)	3.3, 3.4	proof	Figs. 5, 6
Right band	3.6	proof	Fig. 8
Left-border lemma and barrier; no eventually regular half-line	4.1, 4.2, 4.3	proof	Figs. 9–11
Finitely many early rows; the formulas $\mathcal{G}_k(T, N_1, C)$	5.3, 5.5	proof	Figs. 12, 13
Four states excluded by the lines of lengths $2, \ldots, 5$ and parts of those of lengths $6, \ldots, 10$	6.1	proof and certified computation	Fig. 13, Table 3
Sharpness of Theorem 6.1	6.2	exhaustive enumeration; models checked by simulation	Table 3
Five rules for exactly the lengths $2, \ldots, 14$	6.3	proof and exact finite simulation	Figs. 14, 15, Tables 4, 5
At least 15 neighbourhoods on the half-line of a five-state solution	6.4	proof, exhaustive enumeration and certified computation	Fig. 16, Table 5
The half-line of δ_{12} and δ_{13} cannot be completed	6.5	proof and exact finite simulation	Fig. 17
Balzer's conditional five-state result	6.6	certified computation; a model checked by simulation	Table 6
Observations: the onset times $\theta_T(j)$ for Mazoyer's solution and δ_{12}; the patterns of the half-lines of δ_{12} and δ_{14}; PUMP and Lemma 4.1 in finite ranges	3.5, 4, 6.3	finite simulation only	Figs. 6, 7, 14

also exclude minimal-time solutions for variants with sub-generals [18] and for some geometric configurations [19]. Our pumping lemma and our left-border lemma use the same mechanism to constrain solutions of the original problem. That the behaviour behind the front of a cellular automaton is eventually periodic is classical [20]; our contribution is a lower bound on where this periodicity can start. Lower bounds on the number of states are known for restricted classes and for variants such as the ring [21]; see the survey [6]. Satisfiability solvers have been applied to cellular automata before [22], and clausal proofs checked by programs independent of the solvers [23, 24] have certified large computer-assisted results in combinatorics [25, 26]. We are not aware of statements equivalent to Theorems 3.4 and 4.2 and Corollaries 4.3 and 5.3 in the literature.

Organization Section 2 introduces lines, the half-line and their agreement. Section 3 studies the right end of the reflected triangles and proves the late onset of periodicity, Section 4 studies their left end and proves the left-border barrier, and Section 5 reduces their first rows to finitely many cases. Section 6 applies these results to four and five states, and Section 7 discusses open questions. A list of symbols is given at the end. The programs, rule tables and logs behind every computational statement, with scripts that rerun them, are in the supplementary material, as described in the data availability statement at the end.

2 LINES AND THE HALF-LINE

2.1 Rules, Lines and Synchronization

Let S be a finite set of *states* that contains three distinct states: the *quiescent* state L, the *general* G and the *firing* state F. The other states are called *auxiliary*; the rules of this paper use A, B and, for Mazoyer's solution, C. The *working* states are $W := S \setminus \{\mathsf{F}\}$, and $* \notin S$ denotes the border. A *neighbourhood* is a triple $(x, y, z) \in (W \cup \{*\}) \times W \times (W \cup \{*\})$ with $(x, z) \neq (*, *)$, and a *rule* is a map δ from the set of neighbourhoods to S such that

$$\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \delta(\mathsf{L}, \mathsf{L}, *) = \mathsf{L}. \tag{1}$$

If $|S| = k$, there are $k(k-1)k - (k-1) = (k-1)^2(k+1)$ neighbourhoods: 45 for four states, 96 for five and 175 for six.

The *line of length* $n \geq 2$ is the partial map $C_n : \mathbb{N} \times \{1, \ldots, n\} \to S$ given by $C_n(0, 1) = \mathsf{G}$, $C_n(0, i) = \mathsf{L}$ for $2 \leq i \leq n$, and

$$C_n(t+1, i) = \delta\big(C_n(t, i-1),\, C_n(t, i),\, C_n(t, i+1)\big)$$

if the three arguments are defined and none of them is F; otherwise $C_n(t+1, i)$ is undefined. Here $C_n(t, 0) = C_n(t, n+1) = *$ stand for the borders. Thus a cell is defined as long as nothing in its neighbourhood has fired. The rule *synchronizes* the line of length n at time T if $C_n(T, i) = \mathsf{F}$ for $1 \leq i \leq n$ and $C_n(t, i) \in W$ for all $t < T$ and $1 \leq i \leq n$. A *minimal-time solution* is a rule that synchronizes the line of length n at time $2n - 2$ for every $n \geq 2$; it has k states if $|S| = k$.

The *half-line* $C_\infty : \mathbb{N} \times \{1, 2, \ldots\} \to S$ is defined in the same way, with $C_\infty(0, 1) = \mathsf{G}$, $C_\infty(0, i) = \mathsf{L}$ for $i \geq 2$, and a border on the left only. We draw space-time diagrams with time running downwards and the cells from left to right, as in Figure 1. The *front* is the diagonal $t = i - 1$, and the *anti-diagonal* $t + i = m$ is the set of cells (t, i) with $t + i = m$.

2.2 Agreement

Information travels at most one cell per step, so the cells that the general cannot yet have influenced stay quiescent.

Lemma 2.1 (light cone)**.** *For every rule,* $C_n(t, i) = C_\infty(t, i) = \mathsf{L}$ *whenever* $t < i - 1$ *(and* $i \leq n$ *for* C_n*).*

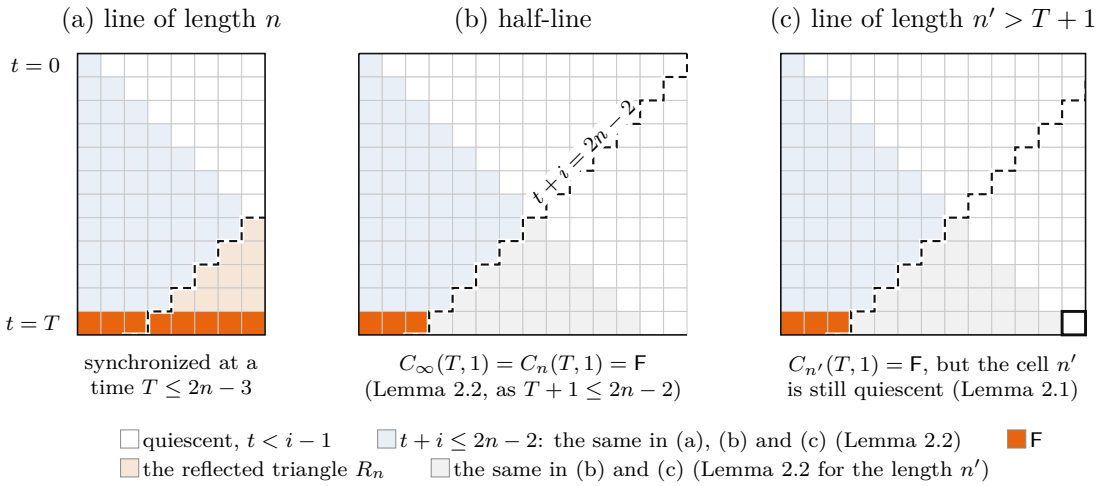


Figure 2: Why no line can be synchronized before time $2n-2$ (Remark 2.3), drawn for $n=8$, $T=11$ and $n'=13$. (a) If the line of length n were synchronized at a time $T \le 2n-3$, its cell $(T,1)$ would satisfy $T+1 \le 2n-2$ and lie above the dashed staircase, where the line agrees with the half-line. (b) Hence the cell $(T,1)$ of the half-line would be in state F. (c) A line of length $n' > T+1$ agrees with the half-line on every cell with $t \le T$, because then $t+i \le T+n' \le 2n'-2$. Its cell 1 would be in state F at time T while its cell n' is still quiescent, so this line could never be synchronized.

Proof. By induction on t. For $t=0$ this is the initial condition. For $t \ge 1$, the cells $(t-1,i')$ with $i' \in \{i-1,i,i+1\}$ satisfy $t-1 < i'-1$, so each of them is L or the right border, and (1) gives $C_n(t,i) = C_\infty(t,i) = \mathsf{L}$. □

The right border of the line of length n is first felt by the cell n at time $n-1$, when the front reaches it, and its influence travels back to the left at speed at most one. Before this echo arrives, the line cannot be told apart from the half-line.

Lemma 2.2 (agreement)**.** *For every rule, every $n \ge 2$, $1 \le i \le n$ and $t \ge 0$ with $t+i \le 2n-2$, the values $C_n(t,i)$ and $C_\infty(t,i)$ are both undefined or both defined and equal.*

Proof. By induction on t; the case $t=0$ is clear. Let $t \ge 1$. If $i=n$, then $t \le n-2 < i-1$, and both values are L by Lemma 2.1. If $i<n$, the cells $(t-1,j)$, $j \in \{i-1,i,i+1\}$, satisfy $(t-1)+j \le 2n-2$ and $j \le n$, so by induction they are both undefined or equal in C_n and C_∞ (for $j=0$ both are the border), and the recursion gives the claim. □

So the border can only matter in the *reflected triangle*

$$R_n := \{(t,i) :\ 1 \le i \le n,\ t+i \ge 2n-1,\ t \le 2n-2\},$$

which is bounded by the anti-diagonal $t+i = 2n-1$, the right border and the firing row $t = 2n-2$ (Figure 1). The agreement lemma already shows why time $2n-2$ is optimal.

Remark 2.3. Suppose that a rule synchronizes every line, the line of length n at some time $T \le 2n-3$. Since $T+1 \le 2n-2$, Lemma 2.2 gives $C_\infty(T,1) = C_n(T,1) = \mathsf{F}$, and then $C_{n'}(T,1) = \mathsf{F}$ for every $n' \ge n$, by Lemma 2.2 again. For $n' > T+1$ the cell n' is still quiescent at time T by Lemma 2.1, so the line of length n' has a cell in state F at time T although not all of its cells are in state F; it can then never be synchronized, a contradiction (Figure 2).

2.3 The Return Chain

The first anti-diagonal of R_n is where the line starts to differ from the half-line, and in a minimal-time solution it differs there everywhere (Figure 1(c) and Figure 3).

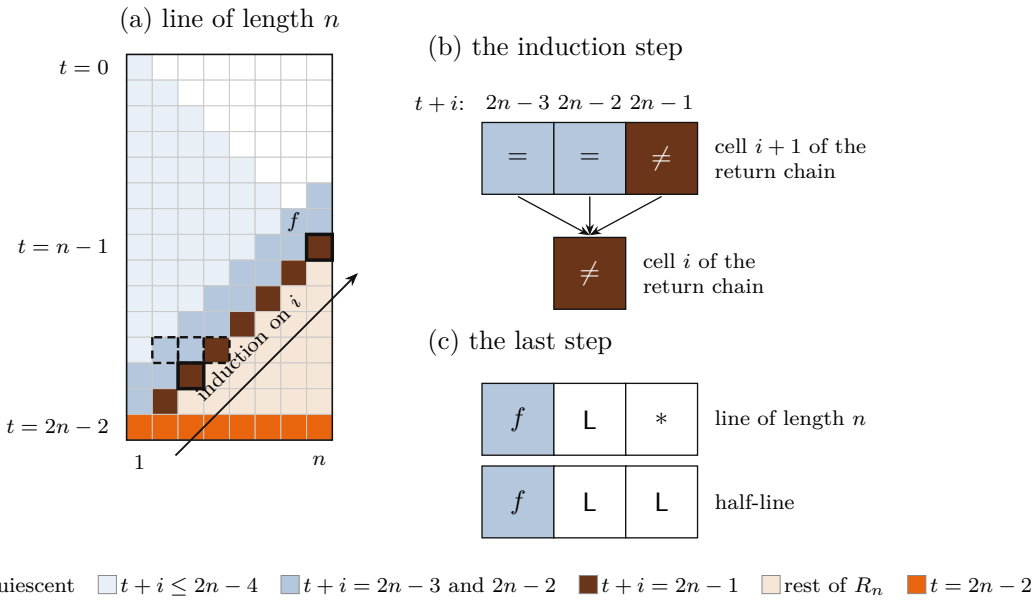


Figure 3: The proof of Lemma 2.4, drawn for $n = 8$. (a) The line of length n. On the cells with $t + i \le 2n - 2$, including the two anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$, it agrees with the half-line (Lemma 2.2). The return chain $t + i = 2n - 1$ starts at the cell $(2n - 2, 1)$, which is F in the line but not in the half-line, and the induction climbs the chain to the cell $(n - 1, n)$. (b) The induction step, outlined in (a) for $i = 3$: the left and middle predecessors of the cell i of the return chain have the same state in the line and in the half-line (=); since the cell differs (≠), so does its right predecessor, the cell $i + 1$ of the return chain. (c) The last cell $(n - 1, n)$ of the return chain, outlined in (a), is computed from f, L, $*$ in the line and from f, L, L in the half-line, where $f = f_{n-1}$ is the front cell; by (1) the results can differ only if $f \neq \mathsf{L}$.

Lemma 2.4 (return chain)**.** *Let $n \ge 2$. If δ synchronizes the lines of lengths n and $n + 1$ at times $2n - 2$ and $2n$, then $C_n(2n - 1 - i, i) \neq C_\infty(2n - 1 - i, i)$ for $1 \le i \le n$, and the front cell $f_{n-1} := C_\infty(n - 2, n - 1)$ is not L.*

Proof. For $i = 1$ we have $C_n(2n - 2, 1) = \mathsf{F}$, whereas $C_\infty(2n - 2, 1) = C_{n+1}(2n - 2, 1) \neq \mathsf{F}$ by Lemma 2.2 (applied to the length $n + 1$) and because $2n - 2 < 2n$. Suppose that the claim holds for some $i < n$. The left and middle predecessors of the cell $(2n - 1 - i, i)$, namely $(2n - 2 - i, i - 1)$ and $(2n - 2 - i, i)$, lie on the anti-diagonals $t + i = 2n - 3$ and $2n - 2$ (for $i = 1$ the left one is the border), so they agree in C_n and C_∞ by Lemma 2.2. As the cell itself differs, so do the right predecessors $(2n - 2 - i, i + 1)$; all these cells are defined, since no cell of the line of length n fires before time $2n - 2$. This is the claim for $i + 1$. For $i = n$ the claim concerns the cell $(n - 1, n)$. Its predecessors satisfy $C_n(n - 2, n - 1) = C_\infty(n - 2, n - 1) = f_{n-1}$ by Lemma 2.2, and the cells $(n - 2, n)$ and $(n - 2, n + 1)$ are quiescent by Lemma 2.1, so the claim reads $\delta(f_{n-1}, \mathsf{L}, *) \neq \delta(f_{n-1}, \mathsf{L}, \mathsf{L})$. This is impossible if $f_{n-1} = \mathsf{L}$, by (1). □

We call the anti-diagonal $t + i = 2n - 1$ the *return chain* of the line of length n. Finally, a minimal-time solution never fires on the half-line: given a cell (t, i), take $n \ge t + i + 1$; then $C_\infty(t, i) = C_n(t, i)$ by Lemma 2.2, and $t < 2n - 2$, so

$$C_\infty(t, i) \in W \qquad \text{for all } t \ge 0 \text{ and } i \ge 1. \tag{2}$$

3 THE RIGHT END

The reflected triangle R_n is computed from the half-line, and its right end fires at time $2n - 2$. We show that the firing cell depends only on a cone of about $n/2$ cells of two anti-diagonals of the half-line. If two lines of different lengths received the same values on this cone, their right

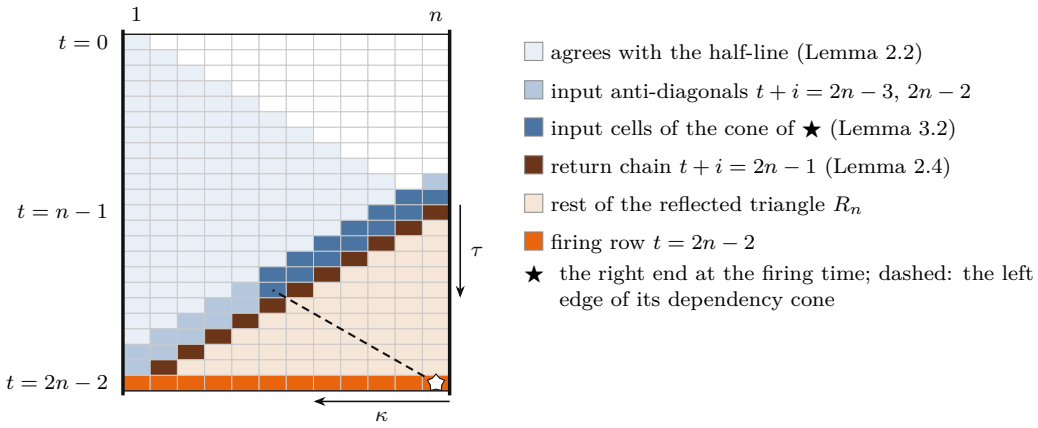


Figure 4: Anatomy of the line of length $n = 12$, coloured by region rather than by state. The upper part agrees with the half-line (Lemma 2.2). The reflected triangle R_n (Lemma 3.1) is computed from the two input anti-diagonals and the borders; its first anti-diagonal is the return chain (Lemma 2.4). The right end at the firing time (★) depends only on the dark input cells (Lemma 3.2); the dashed line is the left edge of its dependency cone. The arrows show the relative coordinates τ and κ of Section 3.

ends would fire at the same relative time, which is impossible (Lemma 3.3). This pumping lemma is a condition on the half-line alone. It implies a lower bound on the times from which the cells behind the front of the half-line are periodic (Theorem 3.4), which we also call the speed-1/3 barrier, for a reason explained after its proof.

3.1 Relative Coordinates and Determinacy

Fix $n \geq 2$ and use the relative coordinates $\tau := t - (n-1)$ and $\kappa := n - i$: the relative time since the front reached the right end, and the distance from the right end. Write $R_n(\tau, \kappa) := C_n(n - 1 + \tau, n - \kappa)$. Then

$$R_n = \{(\tau, \kappa) : \ 0 \leq \kappa \leq \tau \leq n - 1\},$$

the right border is at $\kappa = -1$, the left border at $\kappa = n$, the return chain is the diagonal $\tau = \kappa$, and the firing row is $\tau = n - 1$ (Figure 4). The predecessors of a cell of R_n lie in R_n, on a border, or on one of the anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$, where the line agrees with the half-line (Lemma 2.2). The *input word* of R_n is the partial word

$$J_n(\kappa) := \big(C_\infty(n - 3 + \kappa,\ n - \kappa),\ C_\infty(n - 2 + \kappa,\ n - \kappa)\big), \qquad 0 \leq \kappa \leq n - 1,$$

whose components are the *lower* and *upper input cells*, on the anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$. (For $n = 2$ and $\kappa = 0$ the lower cell lies at time -1; it is a predecessor of no cell of R_n, and we set it to L.)

Lemma 3.1 (determinacy)**.** *There is a partial map Ψ, depending on δ only, such that for all $n \geq 2$ and $(\tau, \kappa) \in R_n$ the values $R_n(\tau, \kappa)$ and $\Psi(J_n, n; \tau, \kappa)$ are both undefined or both defined and equal. Moreover, $\Psi(J, m; \tau, \kappa)$ depends only on the input cells and the borders in the backward dependency cone of (τ, κ).*

In words: the reflected triangle is computed row by row by the local rule, starting from the input word and the two borders, and each of its cells depends only on the input cells from which information can reach it.

Proof. Define $\Psi(J, m; \tau, \kappa)$ for $0 \leq \kappa \leq \tau \leq m - 1$ by induction on τ as $\delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1})$, where each $x_{\kappa'}$ is given by the first applicable case: $*$ if $\kappa' \in \{-1, m\}$; the lower or the upper

component of $J(\kappa')$ if $\tau - 1 = \kappa' - 2$ or $\tau - 1 = \kappa' - 1$, respectively; and $\Psi(J, m; \tau - 1, \kappa')$ otherwise (then $0 \le \kappa' \le \tau - 1$). The value Ψ is undefined if one of the arguments is F or undefined. For $m = n$ the three arguments are the left, middle and right predecessors of the cell $(n - 1 + \tau, n - \kappa)$, and on the input cells C_n and C_∞ are both undefined or equal by Lemma 2.2, so induction on τ gives the first claim. Unfolding the definition gives the second. □

For $(\tau, \kappa) \in R_n$ let $\mathrm{cone}_n(\tau, \kappa)$ be the set of input cells that are reached from (τ, κ) by repeatedly passing to predecessors, stopping at input cells and borders. For the right end at the firing time, $(\tau, \kappa) = (n - 1, 0)$, this cone covers about the right half of the two input anti-diagonals.

Lemma 3.2 (cone of the right end)**.** *For $n \ge 2$, the cone $\mathrm{cone}_n(n - 1, 0)$ avoids the left border. It consists of the upper input cell at $\kappa' = 0$, both input cells at $1 \le \kappa' \le \lfloor n/2 \rfloor$ and, if n is odd, the lower input cell at $\kappa' = (n + 1)/2$.*

Proof. A step to a predecessor decreases τ by one and changes κ by at most one, and the input cells satisfy $\tau' \in \{\kappa' - 1, \kappa' - 2\}$. A path from $(n - 1, 0)$ that reaches an input cell at κ' after r steps therefore has $\kappa' \le r$ and $n - 1 - r \in \{\kappa' - 1, \kappa' - 2\}$, so $2\kappa' \le n + 1$, and $2\kappa' \le n$ for upper cells. Conversely, the upper cell at $\kappa' \le n/2$ is reached by κ' steps to the left followed by vertical steps, and the lower cell at $1 \le \kappa' \le (n + 1)/2$ by $\kappa' - 1$ steps to the left, vertical steps down to $(\kappa' - 1, \kappa' - 1)$ and one more step to the left, through cells of R_n; the lower cell at $\kappa' = 0$ is a predecessor of no cell of R_n. Every cell of R_n on such a path satisfies $\kappa \le \tau \le n - 1 - \kappa$, hence $\kappa \le (n - 1)/2$, so no such cell is adjacent to the left border $\kappa = n$. □

3.2 The Pumping Lemma

Lemma 3.3 (pumping)**.** *Let $2 \le n < n'$, and let δ synchronize the lines of lengths n and n' at times $2n - 2$ and $2n' - 2$. Then J_n and $J_{n'}$ differ on $\mathrm{cone}_n(n - 1, 0)$.*

Proof. Suppose that they agree there. The relative cell $(n - 1, 0)$ lies in $R_{n'}$, and its cone in $R_{n'}$ consists of the same positions, which are away from the left border $\kappa = n'$. By Lemmas 3.1 and 3.2, the relative cell $(n-1, 0)$ of $R_{n'}$ has the value of the relative cell $(n-1, 0)$ of R_n, namely $C_n(2n-2, n) = \mathsf{F}$. Thus $C_{n'}(n' + n - 2, n') = \mathsf{F}$, although $n' + n - 2 < 2n' - 2$ (Figure 5(a)). □

We write $\mathrm{PUMP}(n, n')$ for the conclusion of Lemma 3.3. It is a condition on the half-line alone: the half-line must keep supplying new information to the right ends of longer and longer lines. If the depth rows that carry these cones (defined below) were periodic, the input words of two lengths that differ by a common period would agree on the cone; the next theorem turns this observation into a bound on where the periodicity can start.

3.3 Depth Rows and the Speed-1/3 Barrier

It is convenient to locate the cells of the half-line by their distance behind the front, the cell $(t, t + 1)$. Let

$$\Phi_t(j) := C_\infty(t,\, t + 1 - j) \qquad (0 \le j \le t)$$

be the cell at *depth* j behind the front; the front cell has depth 0. The local rule becomes

$$\Phi_{t+1}(j) = \delta\big(\Phi_t(j),\, \Phi_t(j - 1),\, \Phi_t(j - 2)\big) \qquad (0 \le j \le t + 1), \tag{3}$$

with $\Phi_t(-1) = \Phi_t(-2) = \mathsf{L}$ (the quiescent cells ahead of the front) and $\Phi_t(t + 1) = *$ (the left border). So information only travels to larger depths. The *depth row* j is the sequence $t \mapsto \Phi_t(j)$, $t \ge j$: it consists of the cells $(t, t + 1 - j)$, so it starts in cell 1 at time j and then advances by one cell per step, at distance j behind the front. In these coordinates,

$$J_n(\kappa) = \big(\Phi_{n-3+\kappa}(2\kappa - 2),\ \Phi_{n-2+\kappa}(2\kappa - 1)\big) :$$

the input cells at distance κ from the right end lie on the depth rows $2\kappa - 2$ and $2\kappa - 1$.

(a) the pumping lemma (b) the proof of Theorem 3.4

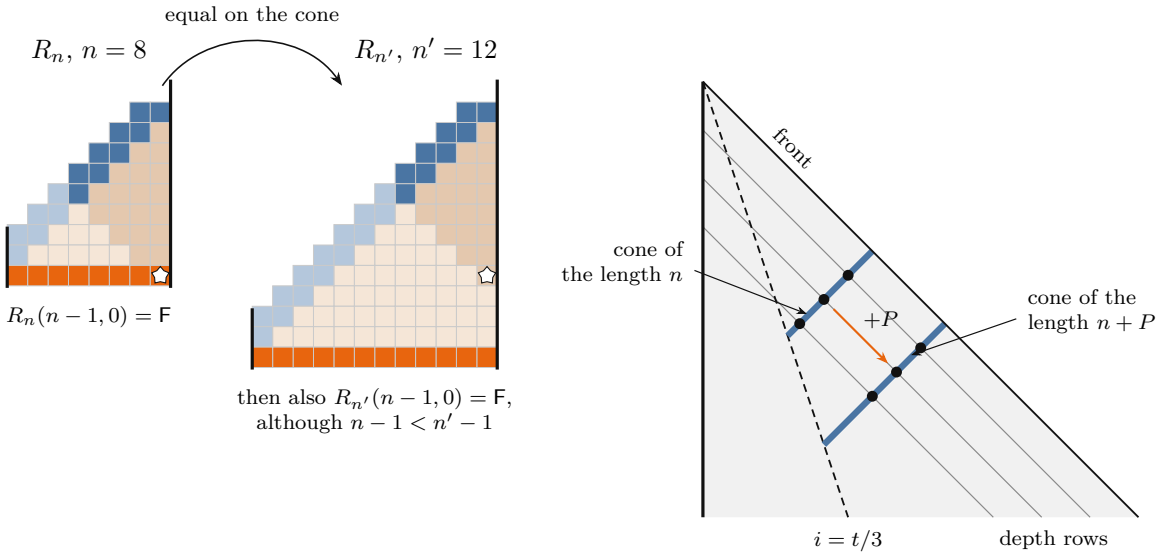


Figure 5: (a) The pumping lemma (Lemma 3.3), drawn for $n = 8$ and $n' = 12$ in relative coordinates: τ runs downwards and κ to the left of the right border (thick line). Above each triangle lie its input cells, on the diagonals $\tau = \kappa - 2$ and $\tau = \kappa - 1$. The right end of R_n at the firing time (★) depends only on the dark input cells (Lemma 3.2), through the darker cells of the triangle. If $R_{n'}$ received the same values at the same input positions, its relative cell $(n-1, 0)$ would also be in state F, although $n-1 < n'-1$. (b) The proof of Theorem 3.4, in the half-line. The input cells of the cone of the right end of the line of length n (thick) lie on the depth rows $0, \dots, n-1$ (grey, parallel to the front), between the front and roughly the line $i = t/3$ (dashed). The cone of the length $n+P$ meets the same depth rows P steps later along each row (arrow). If each of these depth rows were periodic, with a period dividing P, from its point on the cone of n (dots) on, the two cones would carry the same values, which (a) excludes.

Theorem 3.4 (late onset of periodicity)**.** *Let δ be a minimal-time solution. Every depth row $t \mapsto \Phi_t(j)$ is eventually periodic; let $\theta(j)$ be the least $t_0 \geq j$ such that it is periodic on $[t_0, \infty)$. For every $n \geq 4$ there is a depth $j \leq n$ with $\theta(j) > n + j/2 - 5/2$. Hence $\limsup_{j\to\infty} \theta(j)/j \geq 3/2$.*

Proof. Row 0 satisfies $\Phi_{t+1}(0) = \delta(\Phi_t(0), \mathsf{L}, \mathsf{L})$, so it is eventually periodic. By (3), row j is a finite-state system driven by the rows $j-1$ and $j-2$; if these are eventually periodic, so is row j. By (2) all values lie in W. Let P_j be a period of row j, fix $n \geq 4$ and let P be a common multiple of $P_0, \dots, P_{n-1}$.

We compare the input words of R_n and R_{n+P} on $\mathrm{cone}_n(n-1, 0)$. The input cell at $\kappa = 0$ in this cone is the upper one, and it is quiescent in both triangles by Lemma 2.1. The other input cells of the cone have depth $2\kappa - 2$ (lower cells) or $2\kappa - 1$ (upper cells), at most $n-1$ by Lemma 3.2. Suppose that each of them lies in the periodic part of its row (Figure 5(b)), that is, $n - 3 + \kappa \geq \theta(2\kappa - 2)$ for the lower cells and $n - 2 + \kappa \geq \theta(2\kappa - 1)$ for the upper cells. Replacing n by $n + P$ moves each input cell along its depth row by P steps, so $J_{n+P} = J_n$ on the cone, which contradicts Lemma 3.3. Hence some input cell of the cone, of depth $j \leq n$, satisfies $\theta(j) > n - 3 + \kappa$. For a lower cell $\kappa = (j+2)/2$ and for an upper cell $\kappa = (j+1)/2$, so in both cases $\theta(j) > n + j/2 - 5/2$.

Suppose now that $\limsup_{j\to\infty} \theta(j)/j < 3/2$. Then there are $\alpha < 3/2$ and β with $\theta(j) \leq \alpha j + \beta$ for all j; since $\theta(j) \geq j$, we may take $\alpha \geq 1/2$. For all $j \leq n$,

$$\theta(j) - \frac{j}{2} \leq \Big(\alpha - \frac{1}{2}\Big) j + \beta \leq \Big(\alpha - \frac{1}{2}\Big) n + \beta,$$

and since $\alpha - 1/2 < 1$, the right-hand side is at most $n - 5/2$ for n large. Then $\theta(j) \leq n + j/2 - 5/2$ for all $j \leq n$, which contradicts the first part. □

Theorem 3.4 is a statement about the numbers $\theta(j)$, and only about infinitely many of them: it says that $\theta(j) \geq (3/2-\varepsilon)j$ for infinitely many j, and nothing about the other depths. Its geometric reading explains its second name. Depth row j becomes periodic at the cell $(\theta(j), \theta(j)+1-j)$. If $\theta(j) \geq (3/2-\varepsilon)j$, then $j \leq \theta(j)/(3/2-\varepsilon)$, and the position of this cell satisfies

$$\theta(j)+1-j \;\geq\; \frac{1-2\varepsilon}{3-2\varepsilon}\,\theta(j)+1,$$

where the factor $(1-2\varepsilon)/(3-2\varepsilon)$ tends to $1/3$ as $\varepsilon \to 0$ (Figure 6). So, for infinitely many j, depth row j becomes periodic only at a time t at which it is close to the cell $t/3$ or beyond. Equivalently, there is no $c < 1/3$ such that, for all large j, depth row j is periodic from the time at which it reaches the line $i = ct$; this is the sense in which we speak of a speed-$1/3$ barrier. The theorem does not say that the cells that are not yet periodic form a region bounded by a signal, nor that such a boundary moves at speed $1/3$, nor that it exists at all. Remark 3.5 discusses the constructions in which it does.

Remark 3.5 (a heuristic reading). In the classical constructions, the line is divided recursively, and the first division point is where a signal sent by the general meets the echo of the right end (Figure 6(c)); a *signal* of speed v is a pattern that moves by v cells per step. A signal of speed v that leaves the general at time 0 meets the echo, which leaves the right end at time $n-1$ at speed one, at the cell $v(2n-1)/(1+v)$. Suppose that the depth rows become periodic exactly where they cross this signal, as we observed for the two rules simulated below; this is an assumption, not a consequence of minimal time. Then depth row j meets the signal at the time $(j-1)/(1-v)$, so $\theta(j) \approx j/(1-v)$, and Theorem 3.4 gives $v \geq 1/3$ asymptotically: the first division point, at the cell $v(2n-1)/(1+v) \geq (2n-1)/4$, is then asymptotically never in the half of the line next to the general. Waksman's and Balzer's solutions divide the line in the middle, with $v = 1/3$, and Mazoyer's solution divides it at two thirds of its length from the general, with $v = 1/2$.

Simulations can only approximate $\theta(j)$. For a horizon T, let $\theta_T(j)$ be the least $t_0 \geq j$ such that $\Phi_t(j) = \Phi_{t+p}(j)$ for $t_0 \leq t \leq T-p$, where $p \leq 12$ is the least period of depth row j on its last 200 simulated steps; if p is a period of the row from $\theta(j)$ on, then $\theta_T(j) \leq \theta(j)$. For Mazoyer's solution and $T = 2600$, $\theta_T(j) \in \{2j+1, 2j+2\}$ for $1 \leq j < 700$. For the five-state rule δ_{12} of Section 6.3, which synchronizes the lines of lengths $2, \ldots, 12$, and $T = 1400$, $\theta_T(j) = \lceil 3j/2 \rceil + 2$ for $10 \leq j \leq 300$, and $\mathrm{PUMP}(n, n')$ holds for $4 \leq n < n' \leq 650$ (Figures 6 and 7). These are observations on finite horizons, and the reading above need not apply to other rules.

3.4 The Right Band

The pigeonhole argument behind the pumping lemma also works along the right border, in the direction of time. The d cells next to the right border form a finite automaton driven by the column $\kappa = d$, and this automaton has to reach its all-firing state exactly at the firing time.

Lemma 3.6 (right band)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $1 \leq d \leq \tau_0 \leq n-2$ and $\ell \geq 1$, and suppose that the column $\tau \mapsto R_n(\tau, d)$ is ℓ-periodic on $[\tau_0, n-2]$, that is, $R_n(\tau+\ell, d) = R_n(\tau, d)$ whenever $\tau, \tau+\ell \in [\tau_0, n-2]$. Then $n-1-\tau_0 \leq (k-1)^d \ell$.*

Proof. For $d-1 \leq \tau \leq n-1$ put $\mathbf{b}(\tau) := (R_n(\tau,0), \ldots, R_n(\tau, d-1))$. Since $R_n(\tau+1, \kappa) = \delta(R_n(\tau, \kappa+1), R_n(\tau,\kappa), R_n(\tau,\kappa-1))$ with $R_n(\tau,-1) = *$, there is a map Γ, depending on δ and d only, with $\mathbf{b}(\tau+1) = \Gamma(\mathbf{b}(\tau), R_n(\tau,d))$ for $d \leq \tau \leq n-2$; its arguments lie in $W^d \times W$ because no cell fires before time $2n-2$. At $\tau = n-1$ the band is F^d. Suppose that $n-1-\tau_0 > (k-1)^d\ell$. The pairs $(\mathbf{b}(\tau), \tau \bmod \ell)$ with $\tau_0 \leq \tau \leq n-2$ take at most $(k-1)^d\ell$ values, so there are $\tau_0 \leq \tau_1 < \tau_2 \leq n-2$ with $\mathbf{b}(\tau_1) = \mathbf{b}(\tau_2)$ and $\ell \mid h := \tau_2 - \tau_1$. For $0 \leq s \leq n-2-\tau_2$ the driving

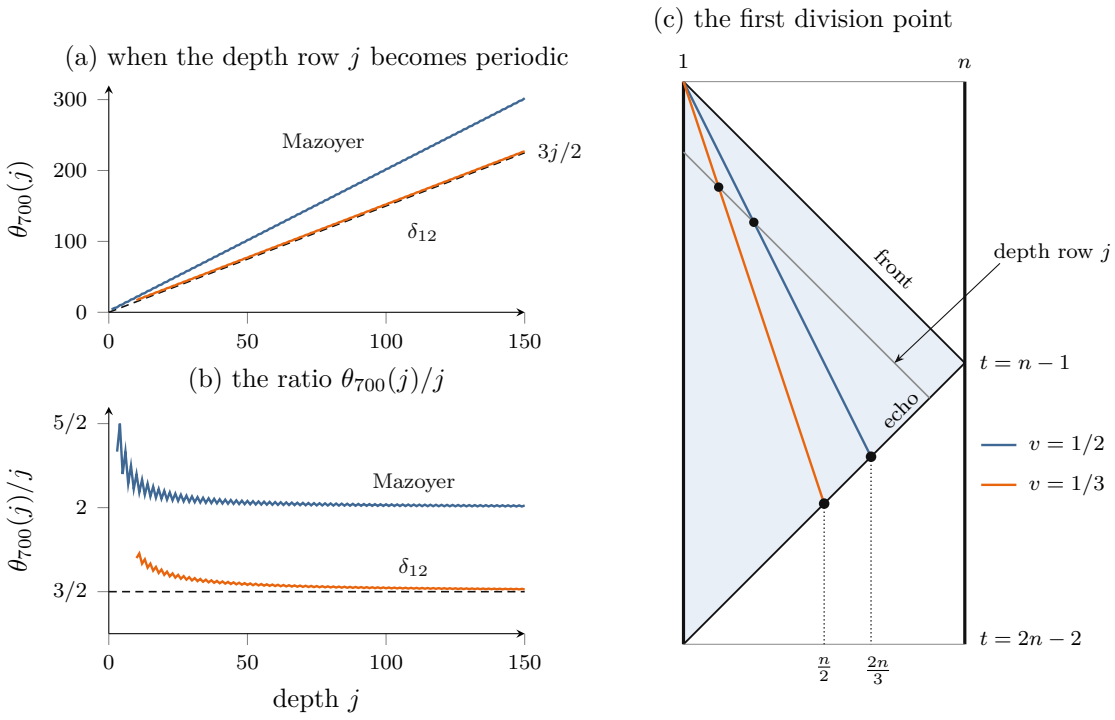


Figure 6: (a) The observed onset $\theta_{700}(j)$ of periodicity of depth row j (Remark 3.5) on the half-lines of Mazoyer's six-state solution ($1 \le j \le 150$) and of the five-state rule δ_{12} ($10 \le j \le 150$), simulated up to time 700; the dashed line is $3j/2$. In this range, $\theta_{700}(j) \in \{2j+1, 2j+2\}$ for Mazoyer's solution and $\theta_{700}(j) = \lceil 3j/2 \rceil + 2$ for δ_{12}. (b) The same data as the ratio $\theta_{700}(j)/j$ (from $j = 3$ on for Mazoyer's solution). Theorem 3.4 says that every minimal-time solution has $\theta(j)/j \ge 3/2 - \varepsilon$ for infinitely many j; the values of δ_{12}, which synchronizes only the lines of lengths $2, \dots, 12$, lie at this bound in the simulated range. (c) The heuristic reading of Remark 3.5 on the line of length n: a signal of speed v sent by the general meets the echo of the right end at the cell $v(2n-1)/(1+v)$, about $n/2$ for $v = 1/3$ and $2n/3$ for $v = 1/2$, and the depth row j (grey) crosses the signal at about the time $j/(1-v)$ (dots). Shaded: the cells reached by the front but not yet by the echo, where the line agrees with the half-line.

cells $R_n(\tau_1 + s, d)$ and $R_n(\tau_2 + s, d)$ lie in $[\tau_0, n-2]$ and their times differ by a multiple of the period, so they are equal, and induction on s gives $\mathbf{b}(\tau_1 + s) = \mathbf{b}(\tau_2 + s)$ for $0 \le s \le n - 1 - \tau_2$. For $s = n - 1 - \tau_2$ this reads $\mathbf{b}(n-1-h) = \mathbf{b}(n-1) = \mathsf{F}^d$, although $n - 1 - h < n - 1$, a contradiction (Figure 8). □

4 THE LEFT END

Near the left border, the reflected triangle is produced by the anti-diagonals $t + i = 2n - 1$, $t + i = 2n$, …, which follow the echo of the right end; we call them *chains*. The first few chains form a finite automaton that reads the two input anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$ and must fire the cells near the left border exactly when the echo reaches them. If the input anti-diagonals were periodic over a long stretch, this automaton would repeat a state and fire too early (Lemma 4.1). Applied to all lengths at once, this excludes periodic wedges at the left border of the half-line (Theorem 4.2).

4.1 Chains and Slices

Fix $n \ge 2$. For $c \ge 0$ and $\max(1, c) \le i \le n$, and for $c = -1$ and $1 \le i \le n - 1$, put

$$v_c(i) := C_n(2n - 2 + c - i,\ i),$$

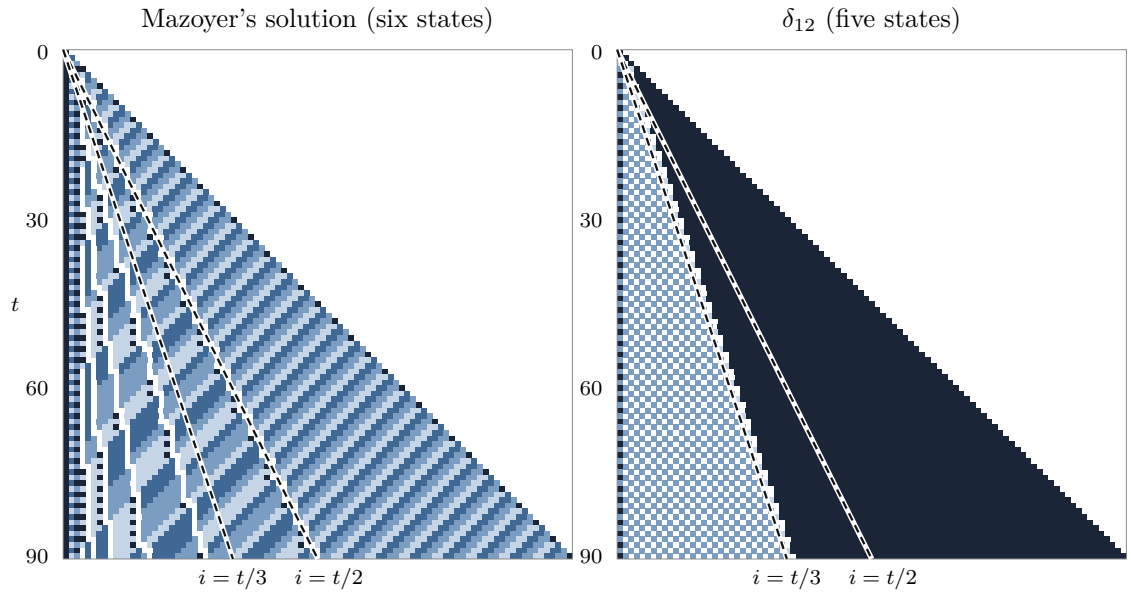


Figure 7: The half-lines of Mazoyer's six-state solution (left) and of the five-state rule δ_{12} (right) for $0 \le t \le 90$. The dashed lines are $i = t/2$ and $i = t/3$. In this range, the depth rows, that is, the cells at a fixed distance behind the front, become periodic near the line $i = t/2$ for Mazoyer's solution and near the line $i = t/3$ for δ_{12} (Remark 3.5). By Theorem 3.4, in a minimal-time solution the depth rows cannot all become periodic before they reach a line $i = ct$ with $c < 1/3$. In the range shown, the half-line of δ_{12} alternates between A and L to the left of the line $i = t/3$, periodically in time and space. Theorem 4.2 excludes such a wedge at the left border if the pattern continues forever, and Corollary 6.5 excludes it for the line of length 518 without this assumption.

the cell at position i of the *chain* $t+i = 2n-2+c$. Chains -1 and 0 are the input anti-diagonals, and chain 1 is the return chain (Figure 9). The predecessors of the cell $(2n-2+c-i, i)$ lie on the chains $c-2$, $c-1$ and c, so

$$v_c(i) = \delta(v_{c-2}(i-1),\, v_{c-1}(i),\, v_c(i+1)) \qquad (c \ge 1), \tag{4}$$

where $v_{c-2}(0)$ and $v_c(n+1)$ stand for the borders. If the line of length n is synchronized at time $2n-2$, then the cell i fires on chain i: $v_i(i) = \mathsf{F}$, and $v_c(i) \ne \mathsf{F}$ for $c < i$. In the skewed coordinates $u_c(y) := v_c(y + \lfloor c/2 \rfloor)$ the recursion (4) becomes

$$u_c(y) = \delta(u_{c-2}(y),\, u_{c-1}(y+\varepsilon_c),\, u_c(y+1)), \qquad \varepsilon_c = 1 \text{ for even } c, \quad \varepsilon_c = 0 \text{ for odd } c, \tag{5}$$

with $u_0(y) = C_\infty(2n-2-y, y)$ and $u_{-1}(y) = C_\infty(2n-2-y, y-1)$. Here $u_c(y)$ is the cell $y + \lfloor c/2 \rfloor$ at time $2n-2+\lceil c/2 \rceil - y$; it is F if $\lceil c/2 \rceil = y$ and not F if $\lceil c/2 \rceil < y$. The *slice* y is the sequence $(u_1(y), u_2(y), \dots)$. Its cells form a staircase that starts just below the input cells $u_{-1}(y)$ and $u_0(y)$, at the cell y of chain 1, and descends to the right: from $u_c(y)$ to $u_{c+1}(y)$ the position increases by one if c is odd, and the time increases by one if c is even. By (5), each slice is computed from the slice $y+1$ and the two input cells $u_{-1}(y)$ and $u_0(y)$: reading the input anti-diagonals from right to left, the slices form a finite automaton. A word w on an integer interval I is *ℓ-periodic* if $w(y+\ell) = w(y)$ whenever $y, y+\ell \in I$.

4.2 The Left-Border Lemma

Lemma 4.1 (left border)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $s, \ell \ge 1$ and $s \le Y \le n-s$, and suppose that the words $y \mapsto C_\infty(2n-2-y, y)$ for $s \le y \le Y$ and $y \mapsto C_\infty(2n-3-y, y)$ for $s \le y \le Y-1$ are ℓ-periodic. Then $Y - s \le (k-1)^{2s}\ell$.*

Proof. For $s \le y \le Y$ let $\mathbf{u}(y) := (u_1(y), \dots, u_{2s}(y))$. These are the cells $y + \lfloor c/2 \rfloor \le n$ at the times $2n-2+\lceil c/2 \rceil - y$ with $\lceil c/2 \rceil \le s \le y$, so they are defined, and none of them is

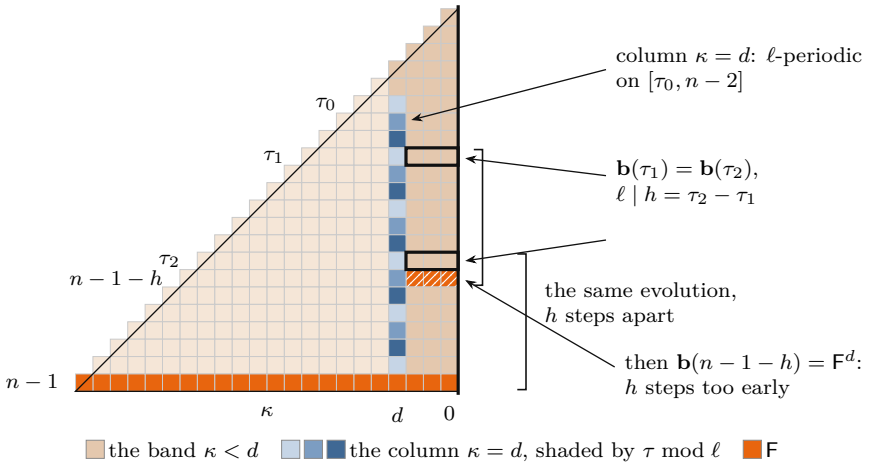


Figure 8: The proof of Lemma 3.6, drawn for $n = 22$, $d = 3$, $\ell = 3$ and $\tau_0 = 5$ in the relative coordinates of R_n. The band of the d cells next to the right border is a finite automaton driven by the column $\kappa = d$, which is ℓ-periodic on $[\tau_0, n-2]$. If the band takes the same value $\mathbf{b}(\tau_1) = \mathbf{b}(\tau_2)$ (outlined) at two times whose difference h is a multiple of ℓ, then it evolves in the same way from τ_1 and from τ_2, so it reaches its all-firing value F^d at the time $n-1-h$ (hatched), h steps before the firing row. As the pairs $(\mathbf{b}(\tau), \tau \bmod \ell)$ take at most $(k-1)^d\ell$ values, this forces $n - 1 - \tau_0 \leq (k-1)^d\ell$.

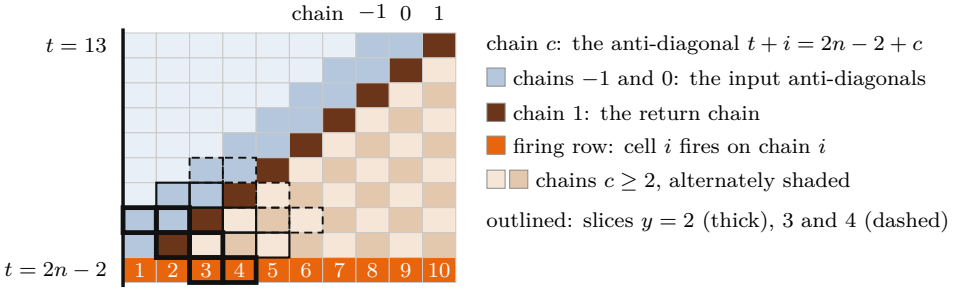


Figure 9: The lower left corner of the line of length $n = 12$, coloured by chain. The cell i of the firing row lies on chain i. Each outlined group of six cells is a slice $(u_1(y), \ldots, u_4(y))$ of the band of chains $1, \ldots, 4$ used in the proof of Lemma 4.1 with $s = 2$, together with its two input cells $u_{-1}(y)$ and $u_0(y)$: the slice is computed from the slice to its right and from these two input cells. The group drawn with a thick line is the one at $y = s$; its last cell $u_{2s}(s)$ is in the firing row.

F when $y > s$. By (5), $u_c(y)$ with $1 \leq c \leq 2s$ is computed from components of index less than c of $\mathbf{u}(y)$, components of index at most c of $\mathbf{u}(y+1)$, and the input cells $u_{-1}(y)$ and $u_0(y)$. Hence $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y))$ for $s + 1 \leq y \leq Y - 1$, with Γ depending on δ only. Unfolding (5) for even c, $u_{2j}(y) = \delta(u_{2j-2}(y), u_{2j-1}(y+1), u_{2j}(y+1))$, shows that $u_{2s}(y) = \Gamma'(\mathbf{u}(y+1), u_0(y))$ for $s \leq y \leq Y - 1$. By hypothesis, u_0 is ℓ-periodic on $[s, Y]$ and u_{-1} on $[s+1, Y]$.

Suppose that $Y - s > (k-1)^{2s}\ell$. The pairs $(\mathbf{u}(y), y \bmod \ell)$ with $s + 1 \leq y \leq Y$ take at most $(k-1)^{2s}\ell$ values, so there are $s + 1 \leq y_1 < y_2 \leq Y$ with $\mathbf{u}(y_1) = \mathbf{u}(y_2)$ and $\ell \mid h := y_2 - y_1$. By downward induction, $\mathbf{u}(y) = \mathbf{u}(y + h)$ for $s + 1 \leq y \leq y_1$: if this holds for $y + 1 \leq y_1$, then

$$\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y)) = \Gamma(\mathbf{u}(y+1+h), u_{-1}(y+h), u_0(y+h)) = \mathbf{u}(y+h),$$

since $y + h < y_2 \leq Y$. As $s + h \leq Y - 1$ and $u_0(s + h) = u_0(s)$,

$$u_{2s}(s+h) = \Gamma'(\mathbf{u}(s+1+h), u_0(s+h)) = \Gamma'(\mathbf{u}(s+1), u_0(s)) = u_{2s}(s) = \mathsf{F},$$

although $\lceil 2s/2 \rceil = s < s + h$, so that $u_{2s}(s + h)$, the cell $2s + h$ at time $2n - 2 - h$, is not F. This is a contradiction. □

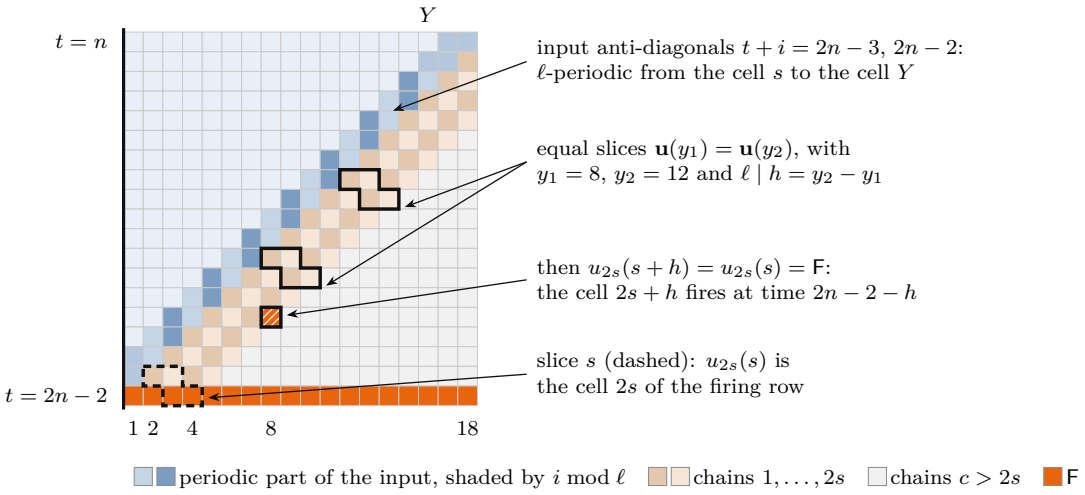


Figure 10: The proof of Lemma 4.1, drawn for $n = 20$, $s = 2$, $\ell = 2$ and $Y = 16$: the lower left corner of the line of length n, at the times $n \le t \le 2n-2$. The two input anti-diagonals $t+i = 2n-3$ and $t+i = 2n-2$ are ℓ-periodic from the cell s to the cell Y. Each slice $\mathbf{u}(y)$, a group of $2s$ cells on the chains $1, \dots, 2s$ (outlined), is computed from the slice $\mathbf{u}(y+1)$ and from two input cells, so the slices form a finite automaton that reads the input anti-diagonals from right to left. If two slices are equal, $\mathbf{u}(y_1) = \mathbf{u}(y_2)$ with $\ell \mid h = y_2 - y_1$, then $\mathbf{u}(y) = \mathbf{u}(y+h)$ for $s+1 \le y \le y_1$, and the last cell $u_{2s}(s+h)$ of the slice $s+h$ (hatched) has the state of the last cell $u_{2s}(s)$ of the slice s, which is F. It is the cell $2s+h$ at the time $2n-2-h$, which would fire h steps too early. Hence $Y - s \le (k-1)^{2s}\ell$.

In words: if the two input anti-diagonals are ℓ-periodic over a stretch of more than $(k-1)^{2s}\ell$ cells that starts at the cell s, then the band of the first $2s$ chains repeats a state, and the cell $2s+h$ fires h steps too early (Figure 10). The bound is exponential in s, but for $s = 1$ and $s = 2$ it is small enough to be useful in computations (Proposition 6.4 and Corollary 6.5).

4.3 No Periodic Wedge at the Left Border

Theorem 4.2 (left-border barrier)**.** *Let δ be a minimal-time solution. There are no $s \ge 1$, $\alpha > 0$, t_0 and $p, q \ge 1$ such that $C_\infty(t+p, i) = C_\infty(t, i)$ and $C_\infty(t, i+q) = C_\infty(t, i)$ whenever the cells involved lie in the wedge*

$$\Omega := \{(t, i) : \ s \le i \le \alpha t, \ t \ge t_0\}.$$

Proof. Suppose otherwise and put $\ell := pq$. If (t, i) and $(t-\ell, i+\ell)$ lie in Ω, so do the cells $(t-\ell, i+jq)$ for $0 \le j \le p$ and $(t-\ell+jp, i)$ for $0 \le j \le q$, hence $C_\infty(t-\ell, i+\ell) = C_\infty(t-\ell, i) = C_\infty(t, i)$. Let $Y_n := \lfloor \alpha(2n-3)/(1+\alpha) \rfloor$. For $s \le y \le Y_n$ the cells $(2n-2-y, y)$ and $(2n-3-y, y)$ satisfy $y \le \alpha(2n-3-y)$, so they lie in Ω as soon as $2n-3-Y_n \ge t_0$, that is, for all large n. A step of ℓ positions along an anti-diagonal is the step from (t, i) to $(t-\ell, i+\ell)$, so the words of Lemma 4.1 are ℓ-periodic on $[s, Y_n]$ and $[s, Y_n - 1]$. As $Y_n \ge 2\alpha n/(1+\alpha) - 4$, for large n the number $Y := \min(Y_n, n-s)$ satisfies $Y - s > (k-1)^{2s}\ell$, which contradicts Lemma 4.1 (Figure 11(a)). □

The half-line of a rule is *eventually regular* if there are t_1, $P \ge 1$ and, for each residue $r \in \{0, \dots, P-1\}$, an integer m, words $w_0, \dots, w_m$, nonempty words $z_1, \dots, z_m$ over W and integers $a_j, D_j \ge 0$ (all depending on r) such that for all $\nu \ge 0$ the configuration $C_\infty(t, 1) \cdots C_\infty(t, t+1)$ at the time $t = t_1 + r + \nu P$ is $w_0 z_1^{a_1+\nu D_1} w_1 \cdots z_m^{a_m+\nu D_m} w_m$. In words: from some time on, the half-line consists of finitely many periodic regions separated by signals of rational speeds; on signals in one-dimensional cellular automata see [20]. The half-line of the five-state rule δ_{12} of Section 6.3 looks like this on all the cells that we simulated, those with $t + i \le 1034$ (Figures 7 and 17): the cells between roughly the cell $t/3$ and the front are in state G, and the

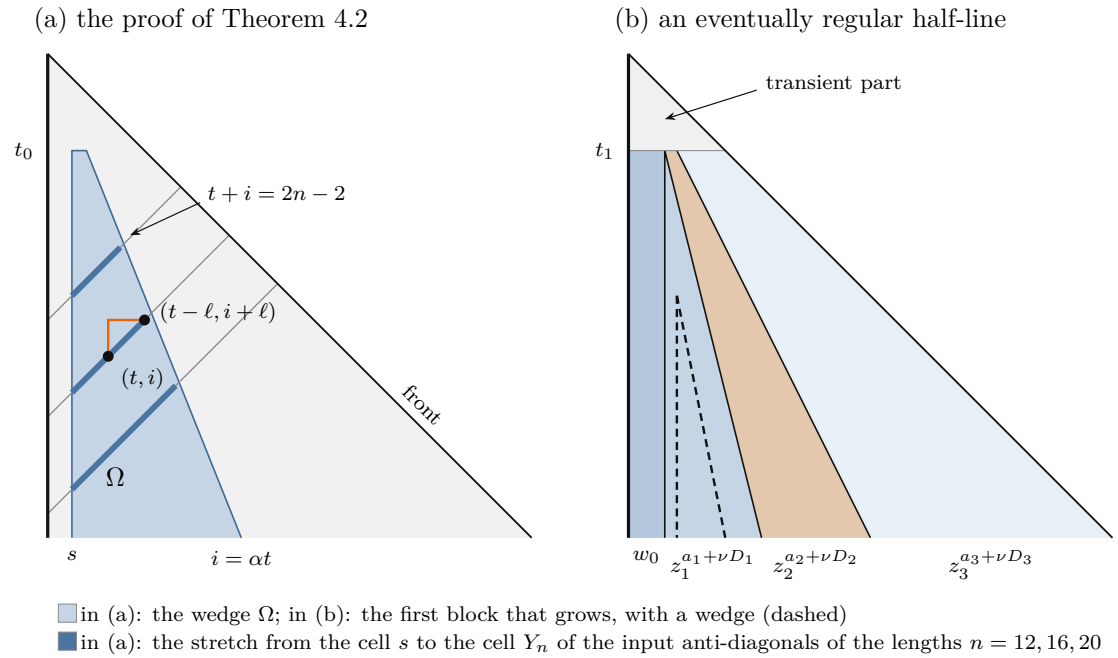


Figure 11: (a) The proof of Theorem 4.2. If the half-line were periodic in time and in space on the wedge $\Omega = \{(t, i) : s \le i \le \alpha t,\ t \ge t_0\}$, then the input anti-diagonals $t + i = 2n - 3$ and $t + i = 2n - 2$ of the line of length n would be ℓ-periodic, with $\ell = pq$, from the cell s to the cell $Y_n \approx 2\alpha n/(1 + \alpha)$ (thick), a stretch whose length grows linearly with n; for large n, Lemma 4.1 gives a contradiction. The path from (t, i) to $(t - \ell, i + \ell)$ shows why: ℓ steps along an anti-diagonal are q time periods p followed by p space periods q inside Ω. (b) An eventually regular half-line (Corollary 4.3): from the time t_1 on, it consists of finitely many periodic blocks separated by signals of rational speeds, here 0, 1/4 and 1/2. The first block that grows contains a wedge (dashed) on which the half-line is periodic in time and space, which (a) excludes.

cells between the left border and roughly the cell $t/3$ alternate between A and L, both in time and in space. Whether it continues in this way is not known, and Corollary 6.5 does not need it.

Corollary 4.3. *The half-line of a minimal-time solution is not eventually regular.*

Proof. Suppose that it is. For each residue r some D_j is positive, because the configuration grows by P cells per period; let $j(r)$ be the least such index. The blocks and the junctions to the left of $z_{j(r)}$ have constant lengths, so the block $z_{j(r)}^{a_{j(r)}+\nu D_{j(r)}}$ starts at a fixed cell s_r, and its length is at least βt for some $\beta > 0$ and all large t of the residue class r. Inside this block, the state of a cell depends only on its distance to s_r modulo $|z_{j(r)}|$, and it is the same at the times t and $t + P$. Let $s := \max_r s_r$, $q := \prod_r |z_{j(r)}|$ and $\alpha := \beta/2$. For t_0 large, every cell of the wedge $\{(t, i) : s \le i \le \alpha t,\ t \ge t_0\}$ lies in the block of index $j(r)$ of its residue class, so C_∞ is periodic with period P in time and q in space on this wedge (Figure 11(b)). This contradicts Theorem 4.2. □

Eventual regularity means exactly the definition above, and Corollary 4.3 excludes nothing beyond it. Theorem 4.2, from which it follows, excludes every half-line that is periodic in time and space on a wedge $\{(t, i) : s \le i \le \alpha t,\ t \ge t_0\}$ with $\alpha > 0$, whatever happens elsewhere. Neither result excludes periodic regions bounded away from the left border, periodic regions at the left border whose width grows sublinearly, or other kinds of regular behaviour that do not produce a periodic wedge at the left border.

5 THE FIRST ROWS AFTER THE REFLECTION

The five-state rules that come closest to minimal time fail in the first rows of their reflected triangles: some cell of a long line fires a few steps after the front has reached the right end (Proposition 6.3 and Figure 14). The condition that this never happens concerns infinitely many lengths. We show that, for a given half-line, it reduces to finitely many lengths, and that a certificate for the reduction can be computed by simulating the half-line.

5.1 Finitely Many Early Rows

For $T \geq 0$ let $E(T) := \{(\tau, \kappa) : 0 \leq \kappa \leq \tau \leq T\}$ be the first $T+1$ rows of a reflected triangle, that is, its cells at the relative times $0, \dots, T$, during the first $T+1$ steps after the front reaches the right end. If $n \geq T+2$, then $E(T) \subseteq R_n$, and no cell of $E(T)$ lies in the firing row $\tau = n-1$.

Lemma 5.1 (early rows)**.** *Let δ be a rule, $T \geq 0$, and $n, n' \geq T+2$ such that the partial values $J_n(\kappa)$ and $J_{n'}(\kappa)$ coincide for $0 \leq \kappa \leq T+1$. Then for every $(\tau, \kappa) \in E(T)$ the values $R_n(\tau, \kappa)$ and $R_{n'}(\tau, \kappa)$ are both undefined or both defined and equal (Figure 12(a)).*

Proof. By Lemma 3.1 we may compare $\Psi(J_n, n; \tau, \kappa)$ and $\Psi(J_{n'}, n'; \tau, \kappa)$. Unfolding the recursion of Lemma 3.1 from $(\tau, \kappa) \in E(T)$ visits only pairs (τ'', κ'') with $0 \leq \kappa'' \leq \tau'' \leq T$ and reads the symbols $x_{\kappa'}$ with $\kappa' \leq \kappa'' + 1 \leq T+1$. The case $\kappa' = m$ would require $m \leq T+1$, which is excluded for $m \in \{n, n'\}$; hence the left border is never read, and the input word is read only at positions $\kappa' \leq T+1$, where J_n and $J_{n'}$ coincide. The two recursions are therefore identical step by step, including the points where an argument is F or undefined. □

The input word on $[0, T+1]$ consists of depth-row cells of depths at most $2T+1$ (Figure 12(b)). These rows are driven only by the rows of smaller depth (3), so the vector of the first $2T+2$ depth rows evolves autonomously, and one period of it is a certificate for all later times.

Lemma 5.2 (periodicity certificate)**.** *Let δ be a rule, $D \geq 0$, and $V_t := (\Phi_t(0), \dots, \Phi_t(D))$ for $t \geq D$. There is a map Γ_D on W^{D+1}, depending on δ and D only, with $V_{t+1} = \Gamma_D(V_t)$ whenever $t \geq D$ and $V_t \in W^{D+1}$. If $V_{t_0} = V_{t_0+p}$ for some $t_0 \geq D$ and $p \geq 1$, and $V_t \in W^{D+1}$ for $t_0 \leq t < t_0 + p$, then $V_t \in W^{D+1}$ and $V_{t+p} = V_t$ for all $t \geq t_0$.*

Proof. For $t \geq D$ and $0 \leq j \leq D$ we have $j \leq t$, so the arguments in (3) are $\Phi_t(j)$, $\Phi_t(j-1)$ and $\Phi_t(j-2)$, with indices at most D, and L for negative indices; none of them is the border symbol $\Phi_t(t+1) = *$. Hence $\Gamma_D(v)_j := \delta(v_j, v_{j-1}, v_{j-2})$, with $v_{-1} = v_{-2} = \mathsf{L}$, has the required property. For the second statement, suppose that $V_{t'} \in W^{D+1}$ and $V_{t'+p} = V_{t'}$ for $t_0 \leq t' \leq t$, which is true for $t = t_0$. Then $V_{t+1+p} = \Gamma_D(V_{t+p}) = \Gamma_D(V_t) = V_{t+1}$, and $V_{t+1} \in W^{D+1}$, by hypothesis if $t+1 < t_0 + p$ and because $V_{t+1} = V_{t+1-p}$ otherwise. Induction on t concludes. □

Corollary 5.3 (finitely many early rows)**.** *Let δ, $T \geq 0$, $D := 2T+1$, $t_0 \geq D$ and $p \geq 1$ satisfy the hypothesis of Lemma 5.2. Then $J_n(\kappa) = J_{n+p}(\kappa) \in W^2$ for all $0 \leq \kappa \leq T+1$ and $n \geq t_0 + 3$. Hence, for every $N_0 \geq T+2$, every R_n with $n \geq N_0$ coincides on $E(T)$, in the sense of Lemma 5.1, with $R_{n'}$ for one of the finitely many lengths $N_0 \leq n' < \max(N_0, t_0+3) + p$.*

Proof. For $1 \leq \kappa \leq T+1$ the components of $J_n(\kappa)$ are cells of the depth rows $2\kappa - 2$ and $2\kappa - 1 \leq D$ at the times $n-3+\kappa$ and $n-2+\kappa$, which are at least $n-2 > t_0$; by Lemma 5.2 they lie in W and do not change when n is replaced by $n+p$. For $\kappa = 0$ the lower component is L, by convention or by Lemma 2.1, and the upper one is $C_\infty(n-2, n) = \mathsf{L}$. If $n \geq \max(N_0, t_0+3) + p$, subtracting a multiple of p gives a length n' in $[\max(N_0, t_0+3), \max(N_0, t_0+3) + p)$ with $J_{n'} = J_n$ on $[0, T+1]$, and Lemma 5.1 applies since $n, n' \geq T+2$. □

For a minimal-time solution, the depth rows are eventually periodic (Theorem 3.4), so t_0 and p as in Corollary 5.3 always exist. Thus the condition that no reflected triangle R_n with $n \geq N_0$ has a firing cell in its first $T+1$ rows reduces to finitely many lengths, with a certificate (t_0, p) found by simulating the half-line.

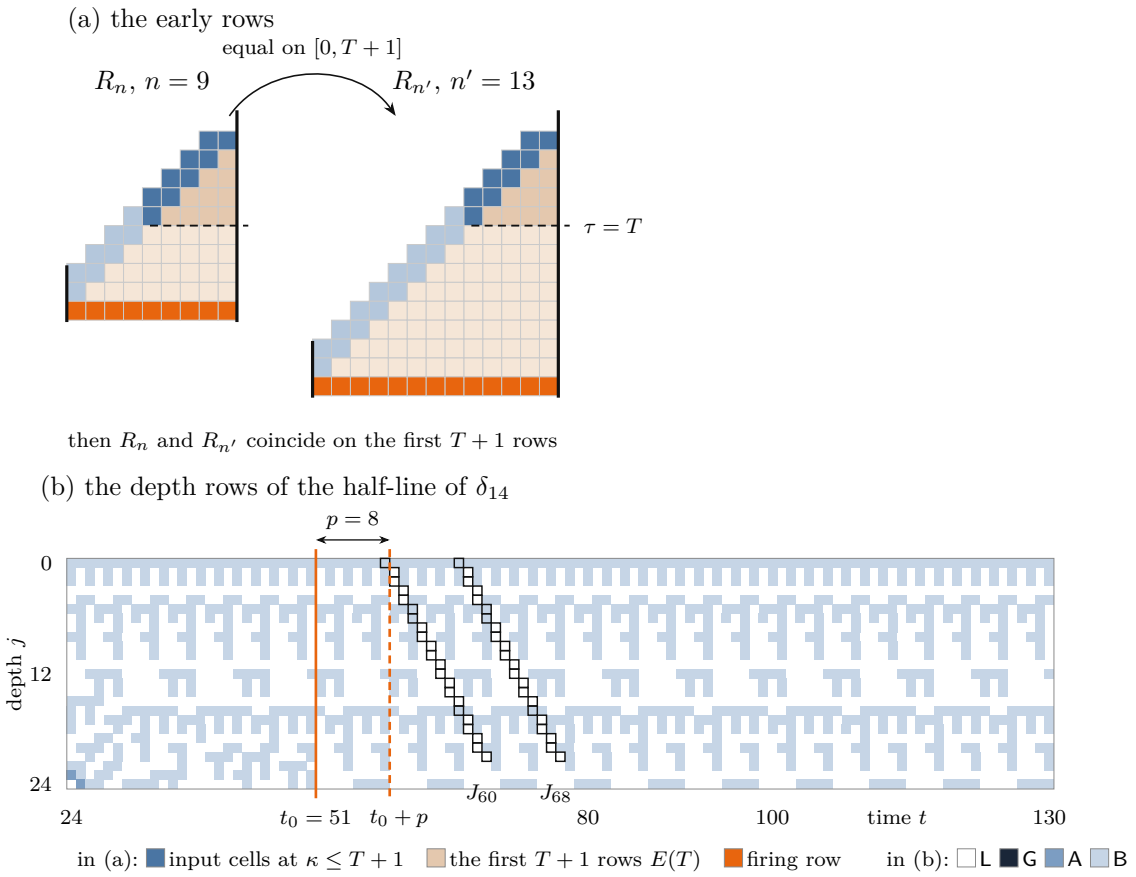


Figure 12: (a) Lemma 5.1, drawn for $T = 3$, $n = 9$ and $n' = 13$ in the relative coordinates of Figure 5(a). The first $T+1$ rows $E(T)$ of a reflected triangle are computed from the right border and the input cells at the distances $\kappa \le T+1$ from it; if these input cells agree, so do the first $T+1$ rows. (b) Lemma 5.2 and Corollary 5.3 for the rule δ_{14} of Section 6.3: the depth rows $j = 0, \dots, 24$ (downwards) of its half-line at the times $t = 24, \dots, 130$ (to the right). From $t_0 = 51$ on, the vector V_t of these rows is periodic with period $p = 8$; the simulation checks $V_t = V_{t+8}$ for $51 \le t \le 152$, and Lemma 5.2 extends it to all $t \ge 51$. Outlined: the input cells $J_n(\kappa)$, $1 \le \kappa \le 11$, of the lengths $n = 60$ and $n = 68 = 60 + p$, on the depth rows $2\kappa - 2$ and $2\kappa - 1$ at times beyond t_0; they carry the same values, as Corollary 5.3 asserts with $T = 10$.

5.2 Early-Row Formulas

For a search over all rules, the half-line is not known in advance, but its first rows and the lines of small lengths can be encoded together. The following conditions collect what a minimal-time solution must satisfy on the complete lines of small lengths, on the early rows and return chains of the next lengths, and on the cone of their right ends.

Definition 5.4. Let $k \ge 3$, $T \ge 1$, $N_1 \ge T+2$ and $T+1 \le C \le N_1$. A k-state rule δ satisfying (1) *satisfies* $\mathcal{G}_k(T, N_1, C)$ if

(G1) δ synchronizes the line of length n at time $2n-2$ for $2 \le n \le T+1$;

(G2) for $T+2 \le n \le N_1$, $1 \le i \le n$ and $t \le n-1+T$: $C_n(t,i) \in W$;

(G3) for $T+2 \le n \le N_1$ and $1 \le i \le n$ with $t := 2n-1-i \le n-1+T$: $C_n(t,i) \ne C_\infty(t,i)$;

(G4) $C_\infty(t,i) \in W$ whenever $t+i \le 2N_1$;

(G5) for $T+2 \le n \le C$: $R_n(\tau,\kappa) \in W$ for all $(\tau,\kappa) \in R_n$ with $\tau+\kappa \le n-1$ and $\tau < n-1$, $R_n(n-1,0) = \mathsf{F}$, and $R_n(\kappa,\kappa) \ne C_\infty(n-1+\kappa, n-\kappa)$ for $2\kappa \le n-1$.

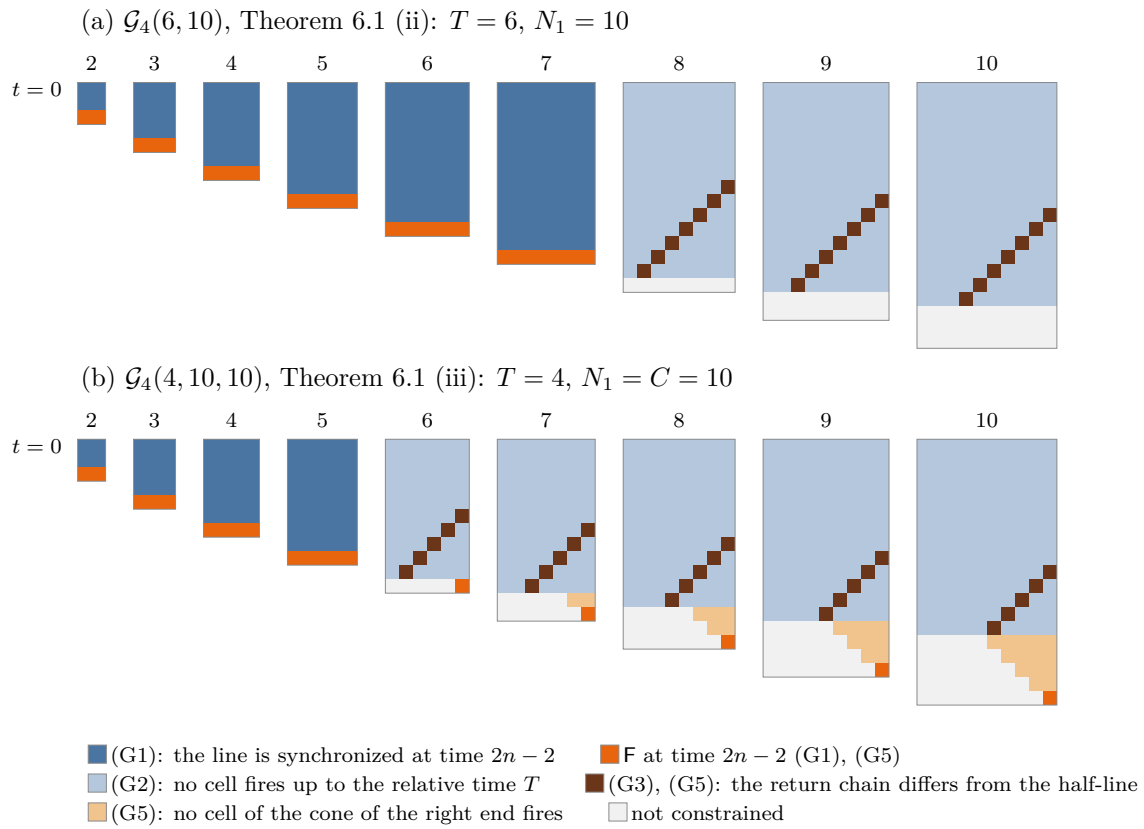


Figure 13: The cells of the lines of lengths $n = 2, \dots, 10$ (time downwards, the line of length n in a column of width n) that the formulas of Theorem 6.1 (ii) and (iii) constrain, coloured by the condition of Definition 5.4 that applies. Condition (G4), on the half-line, is not drawn. (a) In $\mathcal{G}_4(6, 10)$ the lines of lengths $2, \dots, 7$ are complete; for the lengths 8, 9 and 10, only the cells up to the relative time $\tau = 6$ are constrained. (b) In $\mathcal{G}_4(4, 10, 10)$ only the lines of lengths $2, \dots, 5$ are complete; for the lengths $6, \dots, 10$, the formula constrains the cells up to $\tau = 4$, the cone of the right end at the firing time and the return chain in it.

We write $\mathcal{G}_k(T, N_1)$ for $\mathcal{G}_k(T, N_1, T + 1)$, in which (G5) is void.

Figure 13 shows the cells that these conditions constrain in the two formulas of Theorem 6.1. Condition (G1) asks for the complete lines of lengths up to $T + 1$. Condition (G2) says that no cell of the lines of lengths $T + 2, \dots, N_1$ fires up to the relative time $\tau = T$, and (G3) is the return chain of Lemma 2.4, restricted to the relative times $\tau \le T$. Condition (G4) says that the half-line does not fire on the cells with $t + i \le 2N_1$. Condition (G5) concerns the backward dependency cone $\{(\tau, \kappa) \in R_n : \tau + \kappa \le n - 1\}$ of the firing cell $(n - 1, 0)$; by the local definition of C_n, its cells are defined even if a cell outside the cone fires earlier.

Lemma 5.5. *Every k-state minimal-time solution satisfies $\mathcal{G}_k(T, N_1, C)$ for all T, N_1 and C as in Definition 5.4.*

Proof. (G1) holds by definition. For (G2), $n \ge T + 2$ gives $n - 1 + T < 2n - 2$, and every cell of the line of length n is in W before time $2n - 2$. (G3) is Lemma 2.4, which applies because the lines of lengths n and $n + 1$ are synchronized at times $2n - 2$ and $2n$. (G4) is (2). For (G5), the cells of R_n with $\tau < n - 1$ are in W, $R_n(n - 1, 0) = C_n(2n - 2, n) = \mathsf{F}$, and the return chain differs from C_∞ by Lemma 2.4. □

For a fixed half-line, the early rows of all lengths $n \ge N_0$ can thus be checked on the representatives of Corollary 5.3; Proposition 6.3 is proved in this way, and the same representatives allow a formula to impose (G2) and (G3) on all $n \ge N_0$.

6 APPLICATIONS TO FOUR AND FIVE STATES

We now combine the necessary conditions with satisfiability solvers. Each statement of this section rests on one or more of four kinds of evidence, which we state for each result and summarize in Table 1: proofs in the text; exact finite computations, that is, simulations and exhaustive enumerations that decide a finite statement completely; satisfiable formulas, whose models we check by simulation; and unsatisfiable formulas, certified by proofs that an independent checker verifies. Statements that rest on finite simulations alone are observations about the simulated ranges. We mark them as such, and no proof uses them.

6.1 Method

Encodings The formula $\mathrm{MT}_k(\mathcal{N}; M, N')$ (MT for minimal time) has one Boolean variable for each possible value of each entry of a k-state rule table, of each cell of C_∞ with $t + i \le M$, and of each cell of R_n with $n \in \mathcal{N}$; clauses require that exactly one value is chosen for each entry and each cell. Its clauses state the local rule, that no half-line cell is F, that the line of length n is synchronized at time $2n - 2$ for every $n \in \mathcal{N}$, the return chains of Lemma 2.4 for $n, n+1 \in \mathcal{N}$, and $\mathrm{PUMP}(n, n')$ for $4 \le n < n' \le N'$; the formula $\mathrm{MT}_k(\mathcal{N}; M)$ has no pumping clauses. The formula $\mathcal{G}_k(T, N_1, C)$ is encoded in the same way, with $M = 2N_1$. Its regions are closed under predecessors, so the encoding is equivalent to Definition 5.4. Appendix A gives the clauses.

Certificates Every unsatisfiability claim is certified by a proof in the LRAT format [24], which is produced by the solver CaDiCaL [27] or, for one formula, obtained from a DRAT proof of the solver Kissat [28] with `drat-trim` [23], and verified by the independent checker `lrat-check`. The eight unsatisfiable formulas are published with their SHA-256 digests, so that the proofs can be checked against exactly these formulas (Appendix B).

Checks of the encodings A checked proof shows that a formula is unsatisfiable, but not that the formula expresses the intended conditions; the programs that generate the formulas are trusted. We tested them in four independent ways.

(1) *Differential test.* A program written for this purpose from the definitions of this paper, sharing no code with the generators, evaluates the conditions directly by simulating the lines and the half-line: the synchronization of the lines, the half-line up to an anti-diagonal, $\mathrm{PUMP}(n, n')$ with the cone of Lemma 3.2, (G1)–(G5) and (B1)–(B4) of Section 6.5. With a rule table fixed by assumptions, each formula must be satisfiable exactly when the table satisfies its conditions. We compared the two on 34 formulas (those of Theorem 6.1, Proposition 6.2 (b), Propositions 6.4 and 6.6 and Table 2, and variants for four, five and six states) and 3,582 tables (the known rules, models of the satisfiable formulas, the 27 partial rules of Proposition 6.2 (a) completed at random, single and double mutations of these, and random tables): the answers agree on all 57,756 pairs, 973 of which are satisfiable. The formulas of Theorem 6.1 and Proposition 6.6 in this test are identical to the published files (Appendix B), and that of Proposition 6.4 becomes identical to its file when the selector clauses of its 230 germs (Appendix A) are added. The clauses of (B1)–(B4) alone agree with the direct check on 5,200 tables, and the cones found by search agree with Lemma 3.2 for $2 \le n < 200$. No table satisfies the formulas of Theorem 6.1, so for them the test can only confirm that violated conditions are detected; the same generators with other parameters agree with the direct evaluation on satisfiable tables as well.

(2) *Known rules.* With the tables of Mazoyer's solution, as extracted from Duprat's formal proof of its correctness [29], and of δ_{14} fixed, the formulas have the outcomes that

Table 2: Known rules fixed in the formulas (Section 6.1, check (2)): the outcome is predicted by simulating the rule. The formula for all $n \geq 15$ uses the representative lengths of Corollary 5.3.

Table fixed	Conditions	Result	Prediction from simulation
Mazoyer's	$\mathrm{MT}_6(\{3,\dots,12\};78,40)$	satisfiable	synchronizes every line of length $n \geq 3$
Mazoyer's	(G2), (G3) with $T = 8$ for all $n \geq 15$	satisfiable	the same; 21 representative lengths, $(t_0, p) = (42, 3)$
δ_{14}	lines of lengths $2,\dots,14$	satisfiable	synchronizes them
δ_{14}	lines of lengths $2,\dots,15$	unsatisfiable	the line of length 15 fires at $\tau = 8$
δ_{14}	lines $2,\dots,14$ and (G2) for $15,\dots,30$, $T = 6$	satisfiable	no line $15 \leq n \leq 30$ fires at $\tau \leq 6$
δ_{14}	lines $2,\dots,14$ and (G2) for $15,\dots,30$, $T = 8$	unsatisfiable	some line fires at $\tau \in \{7, 8\}$

simulation predicts (Table 2).

(3) *Independent methods.* The formula of Theorem 6.1 (i) and the exhaustive enumeration of Proposition 6.2 (a), a separate program, agree that no four-state rule synchronizes the lines of lengths $2,\dots,9$. The germ enumeration of Proposition 6.4 and the refutations of its 14 remaining germs were each repeated by an independent program.

(4) *Models.* Every model found by a solver was re-simulated by a checker that shares no code with the generators.

These tests make an error in the generators unlikely, but they do not prove its absence; what remains trusted is that the generators, on the formulas of the paper, behave as on the tested ones.

Reproducibility Every computational statement can be repeated with the supplementary material, which is publicly available (see the data availability statement). It contains the programs, the rule tables, the germ lists, the eight unsatisfiable formulas with their SHA-256 digests, the logs, and three scripts: one writes the formulas with the generators and compares them with the recorded digests (about one minute); one reruns the simulations, enumerations, satisfiable formulas and the differential test (about 9 minutes); and one regenerates every LRAT proof and checks it with `lrat-check` (about an hour and 5 GB). CaDiCaL 3.0.1 writes the same proof on every run, and Kissat 4.0.4 with drat-trim reproduced the recorded proof of its case byte for byte, so the regenerated proofs can be compared with the recorded digests. Appendix B lists each computational claim with its programs and files.

6.2 Four States

Theorem 6.1. *No four-state rule satisfying* (1)

(i) synchronizes all lines of lengths $2,\dots,9$ *in minimal time;*

(ii) satisfies $\mathcal{G}_4(6, 10)$*: the lines of lengths* $2,\dots,7$ *complete and, for the lines of lengths* 8, 9 *and* 10*, the first seven rows after the reflection, with the return chains on these rows;*

(iii) satisfies $\mathcal{G}_4(4, 10, 10)$*: the lines of lengths* $2,\dots,5$ *complete and, for the lines of lengths* $6,\dots,10$*, the first five rows after the reflection, with the return chains on these rows, and the right ends.*

In particular, no four-state minimal-time solution exists.

Table 3: The four-state formulas of Theorem 6.1 and Proposition 6.2. The columns (G1), (G2), (G3) and (G5) list the lengths n to which each condition of Definition 5.4 is applied and, for (G2) and (G3), the relative times $\tau \le T$ up to which they are imposed; every formula $\mathcal{G}_4$ also contains (G4). Every unsatisfiability is certified by an LRAT proof that `lrat-check` verifies, and every satisfiable formula has a model that is re-simulated. The labels (i)–(iii) refer to the parts of Theorem 6.1, and (a), (b) to those of Proposition 6.2; the first row is the exhaustive enumeration of part (a), which also finds no partial rule for the lengths $2, \dots, 9$.

Formula	(G1): complete lines	(G2), (G3): early rows and return chains	(G5): right ends	Result
(a) enumeration	$2, \dots, 8$	—	—	27 partial rules
(i) $\mathrm{MT}_4(\{2, \dots, 9\}; 16)$	$2, \dots, 9$	—	—	unsatisfiable
(ii) $\mathcal{G}_4(6, 10)$	$2, \dots, 7$	$8, \dots, 10$; $\tau \le 6$	—	unsatisfiable
(b) $\mathcal{G}_4(5, 30)$	$2, \dots, 6$	$7, \dots, 30$; $\tau \le 5$	—	satisfiable
(b) $\mathcal{G}_4(6, 9)$	$2, \dots, 7$	$8, 9$; $\tau \le 6$	—	satisfiable
(b) $\mathcal{G}_4(6, 10)$ without (G3)	$2, \dots, 7$	$8, \dots, 10$; $\tau \le 6$; (G2) only	—	satisfiable
(iii) $\mathcal{G}_4(4, 10, 10)$	$2, \dots, 5$	$6, \dots, 10$; $\tau \le 4$	$6, \dots, 10$	unsatisfiable
(b) $\mathcal{G}_4(4, 30, 9)$	$2, \dots, 5$	$6, \dots, 30$; $\tau \le 4$	$6, \dots, 9$	satisfiable
(b) $\mathcal{G}_4(3, 30, 12)$	$2, \dots, 4$	$5, \dots, 30$; $\tau \le 3$	$5, \dots, 12$	satisfiable
(b) (G1) and (G5) only	$2, \dots, 5$	—	$6, \dots, 12$	satisfiable

Proof. (i) Such a rule yields a model of $\mathrm{MT}_4(\{2, \dots, 9\}; 16)$: a half-line cell with $t + i \le 16$ is quiescent if $i \ge 10$ (Lemma 2.1) and equals a cell of the line of length 9 at a time $t < 16$ if $i \le 9$ (Lemma 2.2), so it is in W; and the return chains hold by Lemma 2.4 for $2 \le n \le 8$. This formula has 753 variables and 17,605 clauses. (ii) The encoding of $\mathcal{G}_4(6, 10)$ has 919 variables and 22,960 clauses. (iii) The encoding of $\mathcal{G}_4(4, 10, 10)$ has 874 variables and 20,461 clauses. CaDiCaL refutes each of the three formulas with an LRAT proof, and `lrat-check` verifies each proof. The last claim follows from (i), and also from (ii) or (iii) with Lemma 5.5. □

Figure 13 shows the cells that the formulas of parts (ii) and (iii) constrain, and Table 3 compares them with the weaker formulas of the next proposition.

Part (i) certifies the bound observed by Yunès [10]; the earlier proofs [7, 10, 11] rest on the correctness of search programs. Parts (ii) and (iii) show how little information excludes four states: in (iii), only the lines of lengths up to 5 are complete. These bounds are sharp in the following sense.

Proposition 6.2 (computational)**.**

(a) Exactly 27 *partial four-state rules (rules restricted to the neighbourhoods they use) synchronize all lines of lengths* $2, \dots, 8$ *in minimal time, and no partial four-state rule synchronizes all lines of lengths* $2, \dots, 9$*.*

(b) The formulas $\mathcal{G}_4(5, 30)$*,* $\mathcal{G}_4(6, 9)$*,* $\mathcal{G}_4(4, 30, 9)$ *and* $\mathcal{G}_4(3, 30, 12)$*, the formula* $\mathcal{G}_4(6, 10)$ *without (G3), and the formula that consists of (G1) for* $T = 4$ *and (G5) for* $6 \le n \le 12$ *are satisfiable.*

Proof. (a) is an exhaustive enumeration. Yunès [10] found the same number under the extra condition $\delta(*, \mathsf{L}, \mathsf{L}) = \mathsf{L}$, which is vacuous here because none of the 27 partial rules uses the neighbourhood $(*, \mathsf{L}, \mathsf{L})$. (b) The solver finds a model of each formula, and each model is re-simulated. □

In words (Table 3): the formula of part (ii) of Theorem 6.1 becomes satisfiable if one row is removed, even when the lines up to length 30 are added ($\mathcal{G}_4(5, 30)$), if the line of length 10 is removed ($\mathcal{G}_4(6, 9)$), or if the return chains are removed. The formula of part (iii) becomes

Table 4: The five-state rule δ_{14}. The entry in row (x, y) and column z is $\delta_{14}(x, y, z)$; the neighbourhoods $(*, y, *)$ are not used ("–").

x	y	$z=$ L	G	A	B	$*$	x	y	$z=$ L	G	A	B	$*$
$*$	L	F	F	L	G	–	$*$	A	G	G	L	G	–
L	L	L	G	G	B	L	L	A	B	G	B	A	A
G	L	A	B	A	B	G	G	A	A	G	L	A	B
A	L	B	G	G	G	G	A	A	L	A	G	A	B
B	L	B	A	A	L	A	B	A	A	A	G	A	L
$*$	G	L	F	L	L	–	$*$	B	G	G	G	G	–
L	G	A	F	B	A	F	L	B	B	G	A	L	G
G	G	B	F	G	F	B	G	B	B	A	B	B	F
A	G	A	G	G	A	B	A	B	B	B	G	B	G
B	G	A	G	A	L	B	B	B	L	A	A	L	A

satisfiable if the right end of the line of length 10 is removed, even when the early rows of the lines up to length 30 are added ($\mathcal{G}_4(4, 30, 9)$), or if one row is removed, even when the right ends of the lines up to length 12 are added ($\mathcal{G}_4(3, 30, 12)$). The right ends alone, without the early rows and the return chains, do not suffice either.

6.3 Five States: Rules for the Lengths Up to Fourteen

The following five-state rules were found with satisfiability solvers. Their tables are the files `uniformA_2-12.txt` (the rule δ_{12}), `delta13.txt` and `delta14*.txt` in `results/` of the supplementary material. Table 4 shows δ_{14}, and Table 5 summarizes what we know about the seven rules.

Proposition 6.3. *Each of the five-state rules δ_{14}, δ_{14}^{b}, δ_{14}^{c}, δ_{14}^{d} and δ_{14}^{e} synchronizes the line of length n at time $2n-2$ for $2 \le n \le 14$, and for no $n \ge 15$. More precisely, for every $n \ge 15$, some cell of the line of length n is* F *at a relative time $\tau \le \min(T, n-2)$, where $T = 10, 14, 22, 12, 14$, respectively.*

Proof. The lines of lengths $2, \ldots, 60$ are simulated. For $15 \le n \le 60$ some cell is F at a relative time $\tau \le n-2$, which proves the claim for $15 \le n < T+3$. The half-line has no cell in state F at the times $t < 3000$, so by Lemma 2.2 a cell of a line of length $n \le 1500$ that is F at a time $t \le 2n-2$ lies in R_n. Simulating the half-line gives t_0 and p that satisfy the hypothesis of Lemma 5.2 for $D = 2T+4 \ge 2T+1$, namely $(t_0, p) = (51, 8)$, $(66, 3)$, $(257, 192)$, $(95, 48)$ and $(56, 2)$. For every n with $\max(15, T+3) \le n < t_0 + 3 + p \le 452$, direct simulation shows that the line of length n, or of a length n' in this range with $J_{n'} = J_n$ on $[0, T+1]$, has a cell in state F at a relative time $\tau \le T$, hence in $E(T)$. (One length is simulated for each class of a partition that is finer than equality of J_n on $[0, T+1]$; together with the lengths below $T+3$, these are 35, 39, 413, 116 and 28 lengths, the largest being 49, 53, 427, 130 and 42.) By Lemma 5.1 and Corollary 5.3 with $N_0 = \max(15, T+3)$, every R_n with $n \ge N_0$ has a cell of $E(T)$ in state F, and there $\tau \le T \le n-3$ (Figure 15 and Table 5). □

For example, δ_{14} synchronizes the line of length 14 at time 26, whereas on the line of length 15 the cells 13, 14 and 15 fire at time 22, eight steps after the front has reached the right end and six steps too early (Figure 14). The following are observations from simulations, valid in the stated ranges. For $3 \le t < 1500$ the half-line of δ_{14} takes the values L, G at cell 1, L, A at cell 2 and L, B at the cells $i \ge 3$, and δ_{14} restricted to $\{\mathsf{L}, \mathsf{B}\}^3$ is the elementary rule 22 [30]. Neither barrier excludes this half-line in the ranges we checked: PUMP(n, n') holds for $4 \le n < n' \le 800$, and Lemma 4.1 does not apply for $n \le 5000$, $s \le 3$ and $\ell \le 24$. The

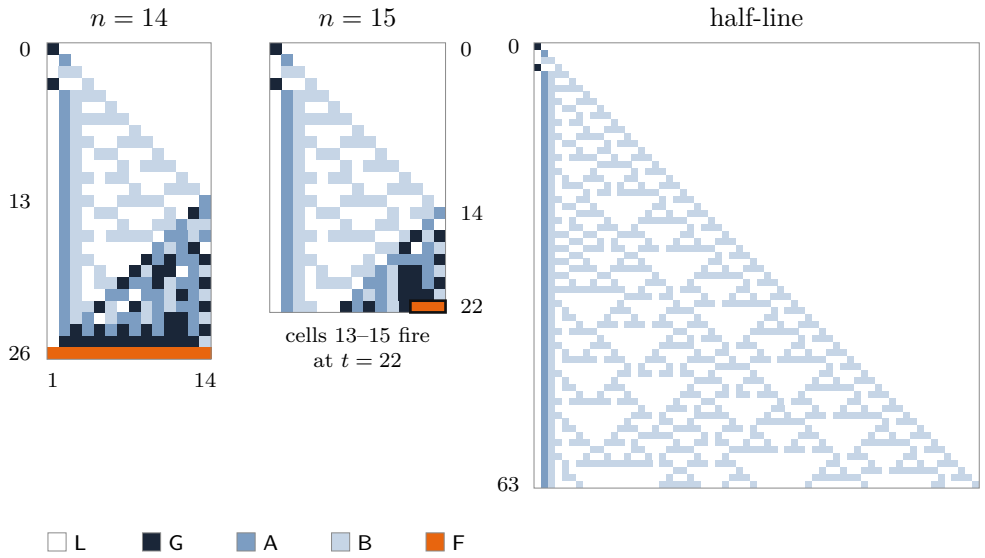


Figure 14: The rule δ_{14}. Left: the line of length 14 is synchronized at time 26. Middle: on the line of length 15, the cells 13, 14 and 15 fire at time 22, that is, at the relative time $\tau = 8$, six steps early. Right: the half-line for $0 \le t \le 63$. In this range, and in our simulations up to $t < 1500$, the cells $i \ge 3$ take only the values L and B from time 3 on, and there δ_{14} acts as the elementary cellular automaton 22.

Table 5: The seven five-state rules of Section 6.3, with their files in `results/` of the supplementary material. Synchronized: the lengths n whose line is synchronized at time $2n - 2$, among $2 \le n \le 60$ for δ_{12} and δ_{13} and among all $n \ge 2$ for the five rules of Proposition 6.3. For these five rules: the bound T of the proposition, the certificate (t_0, p) of Lemma 5.2 for $D = 2T + 4$, and the number of lengths simulated in the proof, with the largest of them. F at $\tau \le 9$: the least length whose line has a cell in state F at a relative time $\tau \le 9$. Last column: the complexity c_{78} of Section 6.4.

Rule	File	Synchronized	T	(t_0, p)	Simulated lengths	F at $\tau \le 9$	c_{78}
δ_{12}	`uniformA_2-12`	$2, \dots, 12$				$n = 16$	22
δ_{13}	`delta13`	$2, \dots, 13$				$n = 14$	22
δ_{14}	`delta14`	$2, \dots, 14$	10	$(51, 8)$	35, up to 49	$n = 15$	16
δ_{14}^{b}	`delta14b`	$2, \dots, 14$	14	$(66, 3)$	39, up to 53	$n = 19$	16
δ_{14}^{c}	`delta14c`	$2, \dots, 14$	22	$(257, 192)$	413, up to 427	$n = 15$	15
δ_{14}^{d}	`delta14d`	$2, \dots, 14$	12	$(95, 48)$	116, up to 130	$n = 15$	15
δ_{14}^{e}	`delta14e`	$2, \dots, 14$	14	$(56, 2)$	28, up to 42	$n = 16$	17

following are exact finite computations. The rules δ_{12} and δ_{13} synchronize the lines of lengths $2, \dots, 12$ and $2, \dots, 13$, respectively, and no other line of length at most 60. Each of the seven rules has a line of some length $14 \le n \le 19$ with a cell in state F at a relative time $\tau \le 9$, so none of them satisfies $\mathcal{G}_5(9, N_1)$ for $N_1 \ge 19$.

What is new Synchronizing finitely many lengths is not difficult in itself: for every k there is a largest $N_k \in \{1, 2, \dots\} \cup \{\infty\}$ such that some k-state rule synchronizes the lines of lengths $2, \dots, N_k$ in minimal time, and since there are finitely many rules, $N_k = \infty$ exactly when a k-state minimal-time solution exists. The question is how far N_k reaches. Exhaustive searches give $N_4 = 8$ (Yunès [10]; Proposition 6.2 (a)), and six-state minimal-time solutions exist [9]. Proposition 6.3 gives $N_5 \ge 14$. We are not aware of an earlier published bound for five states: the five-state constructions we know of synchronize restricted families of lengths, some of them in non-minimal time [14, 15]. The rules add three things to the method of this paper. First, a proof that five states do not suffice must use lines of length at least 15, or conditions other than the synchronization of the lines of lengths $2, \dots, 14$. Second, the statement that each rule fails

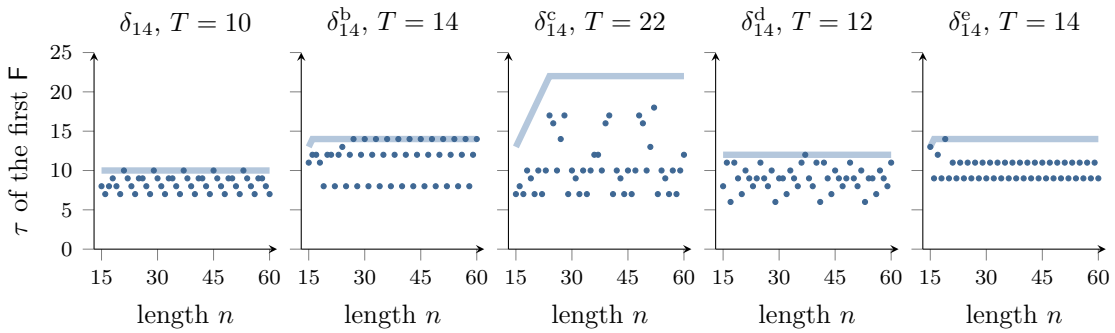


Figure 15: The relative time $\tau = t - (n-1)$ of the first cell in state F on the line of length n, $15 \le n \le 60$, for the five rules of Proposition 6.3 (dots, from direct simulation). The thick line is the bound $\min(T, n-2)$ of the proposition. Every dot lies on or below it, so each of these lines fires before time $2n-2$, at a relative time $\tau \le T$. For $n > 60$ the proposition rests on the certificates of Table 5 and on Corollary 5.3, not on this figure.

for every $n \ge 15$ is not a finite simulation: it combines the simulation of finitely many lengths with Corollary 5.3. Third, the rules fail in the first rows after the reflection, which is what the formulas of Section 5 target. They are not candidates for minimal-time solutions.

6.4 Five States: Half-Lines of Low Complexity

The barriers constrain the half-line, so it is natural to ask how simple the half-line of a five-state minimal-time solution can be. We measure this by the number of distinct neighbourhoods that it uses in a fixed region. The *germ* below M of a rule is its restriction to the neighbourhoods of the cells (t, i) of its half-line with $t \ge 1$ and $t + i \le M$, and its *complexity* $c_M(\delta)$ is the number of these neighbourhoods other than $(\mathsf{L}, \mathsf{L}, \mathsf{L})$. We use $M = 78$: the half-line below the anti-diagonal $t + i = 78$ contains the input anti-diagonals of all lines of lengths $n \le 40$, so the germ below 78 determines the conditions $\mathrm{PUMP}(n, n')$ for $n' \le 40$. For example, $c_{78} = 57$ for Mazoyer's solution, 22 for δ_{12} and 16 for δ_{14} (Table 5).

Proposition 6.4. *Every five-state minimal-time solution δ satisfies $c_{78}(\delta) \ge 15$.*

Proof. Suppose that $c_{78}(\delta) \le 14$. Exchanging A and B if necessary, we may assume that A occurs before B (if at all) in the order of anti-diagonals and, within an anti-diagonal, of time. By (2) and Lemmas 2.4 and 3.3, the germ of δ below 78 produces no F, never produces L at the front, and satisfies $\mathrm{PUMP}(n, n')$ for $4 \le n < n' \le 40$. A depth-first search that chooses the value of each neighbourhood at its first occurrence enumerates all such germs, visiting about 10^8 nodes: there are 244, and an independent re-implementation finds the same 244. For 230 of them, the formula $\mathrm{MT}_5(\{2, \dots, 10\}; 78, 39)$ with the germ fixed is unsatisfiable. A single formula covers all 230: it has a selector variable for each germ, clauses that make each selector fix its germ, and a clause that requires some selector; CaDiCaL refutes it with an LRAT proof that `lrat-check` verifies. Each of the other 14 germs determines C_∞ up to the anti-diagonal 3300, and there Lemma 4.1 applies with $s = 1$: for some $n \le 163$ and $\ell \le 5$, the upper input anti-diagonal of the line of length n is ℓ-periodic on $[1, Y]$ and the lower one on $[1, Y-1]$ with $Y - 1 > 16\ell$ (for example $n = 28$, $\ell = 1$ and $Y = 19$, Figure 16(b)), as an independent simulation confirms. Hence no rule that contains one of the 244 germs is a minimal-time solution (Figure 16(a)). □

The bound is attained by rules that come close to minimal time: the germs below 78 of δ^{c}_{14} and δ^{d}_{14} have complexity 15, and both rules synchronize the lines of lengths $2, \dots, 14$ (Proposition 6.3).

The rules δ_{12} and δ_{13} have the same half-line on the cells with $t + i \le 78$, where they use the 22 neighbourhoods

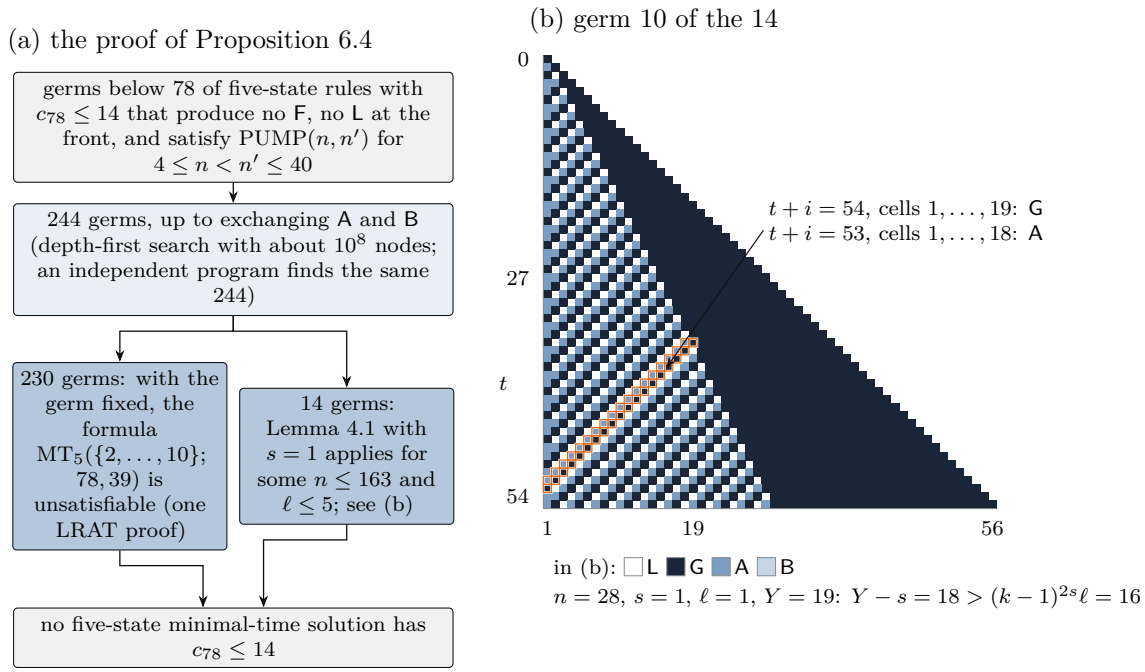


Figure 16: (a) The case analysis in the proof of Proposition 6.4. (b) One of the 14 germs that the formula does not refute: the half-line of germ number 10 of the file `germs_K14_surv10.txt` for $0 \le t \le 55$. For $n = 28$, the upper input anti-diagonal $t + i = 2n - 2 = 54$ is in state G on the cells $1, \dots, 19$, and the lower one, $t + i = 53$, is in state A on the cells $1, \dots, 18$ (outlined). So the hypothesis of Lemma 4.1 holds with $k = 5$, $s = 1$, $\ell = 1$ and $Y = 19$, and $Y - s = 18 > 16 = (k-1)^{2s}\ell$: no rule that contains this germ synchronizes the line of length 28 in minimal time.

```
*GL>A GLL>G *AG>B AGL>L *BL>A BLG>L LGL>B *AL>G ALB>A LBG>L BGL>G
*GA>A GAL>L ALG>B LGG>G GGL>G BGG>L GGG>G ALL>A LLG>L ALA>A LAL>L
```

besides (L, L, L); here `xyz>d` means $\delta(x, y, z) = d$. On these cells, the cells between roughly the cell $t/3$ and the front are in state G, and the cells between the left border and roughly the cell $t/3$ alternate between A and L, periodically in time and space (Figure 7). If this pattern continued forever, Theorem 4.2 would exclude the half-line; Lemma 4.1 gives an explicit length without that assumption.

Corollary 6.5. *No five-state rule that has the* 22 *transitions above synchronizes the line of length* 518 *in minimal time.*

Proof. With these transitions and $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$, no other neighbourhood occurs on the cells with $t + i \le 1034$, so every such rule has the same diagram C_∞ there. On the anti-diagonal $t + i = 1034$ the cells $2, \dots, 259$ are in state A, and on $t + i = 1033$ the cells $2, \dots, 258$ are in state L. Lemma 4.1 with $k = 5$, $n = 518$, $s = 2$, $\ell = 1$ and $Y = 259$ would give $257 \le 4^4 = 256$ (Figure 17). □

6.5 Balzer's Conditions

Balzer [7, p. 37] reported that no five-state minimal-time solution satisfies the following four conditions:

(B1) $\delta(x, \mathsf{G}, y) = \mathsf{G}$ unless $x, y \in \{\mathsf{G}, *\}$;

(B2) $\delta(I(z), I(y), I(x)) = I(\delta(x, y, z))$ for an involution I of the states with $I(*) = *$;

(B3) $\delta(\mathsf{G}, \mathsf{G}, \mathsf{G}) = \delta(*, \mathsf{G}, \mathsf{G}) = \delta(\mathsf{G}, \mathsf{G}, *) = \mathsf{F}$, and no other neighbourhood gives F;

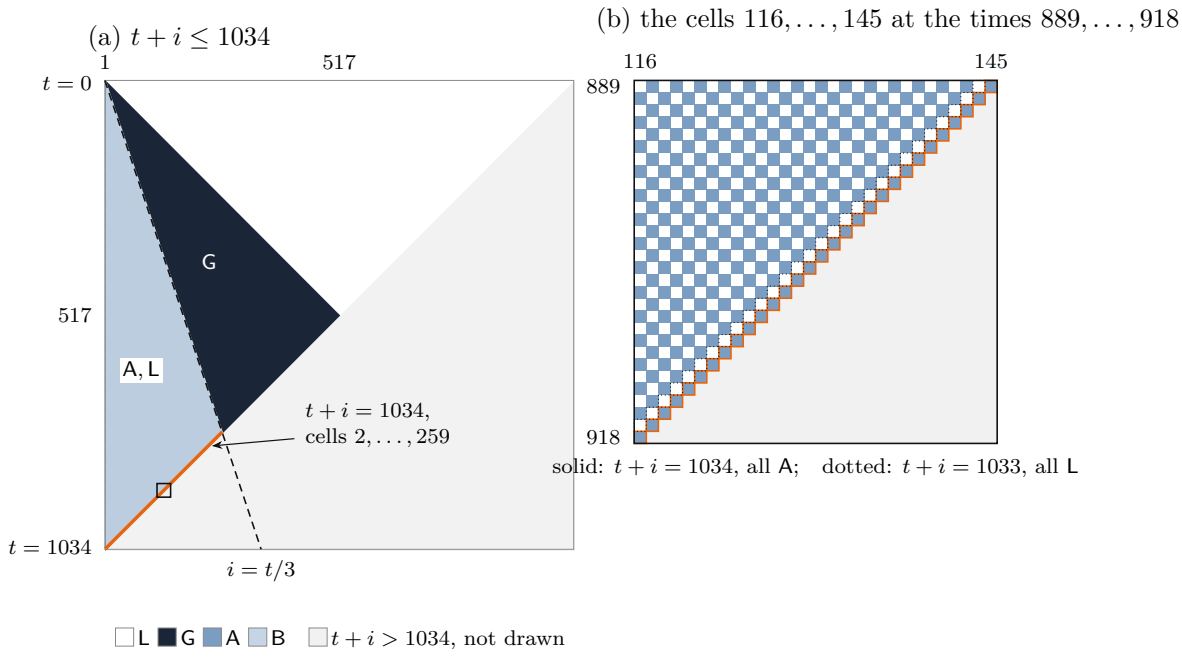


Figure 17: The half-line of δ_{12} on the cells with $t + i \le 1034$, which every rule with the 22 transitions of Corollary 6.5 shares. (a) All these cells, one pixel per cell: the region in state G (dark) lies between roughly the line $i = t/3$ (dashed) and the front, and the cells between the left border and roughly the line $i = t/3$ alternate between A and L. The thick segment marks the cells $2, \dots, 259$ of the anti-diagonal $t + i = 1034$. (b) The cells in the small square of (a). On $t + i = 1034$ (solid outlines) they are all in state A, and on $t + i = 1033$ (dotted outlines) all in state L. This holds for the cells $2, \dots, 259$ and $2, \dots, 258$, respectively, so Lemma 4.1 applies to the line of length 518 with $s = 2$, $\ell = 1$ and $Y = 259$.

(B4) $\delta(\mathsf{G}, v, \mathsf{G}) = \mathsf{G}$ for all $v \in W \setminus \{\mathsf{G}\}$.

In the *weak reading* [10], (B1) is required only for working states x, y, and (B3) without its last clause. We take $I(\mathsf{F}) = \mathsf{F}$. Then $I(\mathsf{G}) = \mathsf{G}$: otherwise $g := I(\mathsf{G}) \in W \setminus \{\mathsf{G}\}$, and (B2) and (B4) give $\delta(g, \mathsf{G}, g) = I(\delta(\mathsf{G}, g, \mathsf{G})) = I(\mathsf{G}) = g$, whereas (B1) gives $\delta(g, \mathsf{G}, g) = \mathsf{G}$. So I is one of the four involutions of $\{\mathsf{L}, \mathsf{A}, \mathsf{B}\}$, and (B1)–(B4) become unit and binary clauses.

Proposition 6.6. *No five-state rule that satisfies (B1)–(B4) in the weak reading, with $I(\mathsf{F}) = \mathsf{F}$, synchronizes all lines of lengths $2, \dots, 12$ in minimal time; a fortiori, no rule in Balzer's form does. The lengths up to 11 do not suffice to prove this.*

Proof. For each involution, the clauses of (B1)–(B4) are added to $\mathrm{MT}_5(\{2, \dots, N\}; 2N - 2)$. For $I = (\mathsf{A}\,\mathsf{B})$ and $N = 10$, and for $I = (\mathsf{L}\,\mathsf{A})$ and $I = (\mathsf{L}\,\mathsf{B})$ and $N = 9$, CaDiCaL refutes the formula with an LRAT proof. For I the identity and $N = 12$, Kissat refutes it; drat-trim turns its DRAT proof into an LRAT proof. Each of the four proofs is verified by `lrat-check`. The rule `results/balzer_sym11.txt` satisfies (B1)–(B4) in the strong reading with I the identity and synchronizes every line of length at most 11 (direct simulation); see Table 6. □

7 DISCUSSION

The conditions of this paper are of two kinds. The barriers of Sections 3 and 4 are proved for every number of states and concern the half-line alone. The conditions of Section 5 concern the first rows after the reflection and the right ends, and we used them through satisfiability solvers, for four and five states.

Table 6: The formulas of Proposition 6.6: the clauses of (B1)–(B4) for the involution I added to $\mathrm{MT}_5(\{2,\dots,N\};2N-2)$, where $2,\dots,N$ are the lengths in the second column. Every proof is verified by `lrat-check`. The last row is a rule, found by a solver and checked by direct simulation, that satisfies the strong reading, hence the weak one.

Involution I	Lengths	Reading	Result	Certificate
(A B)	$2,\dots,10$	weak	unsatisfiable	LRAT proof of CaDiCaL
(L A)	$2,\dots,9$	weak	unsatisfiable	LRAT proof of CaDiCaL
(L B)	$2,\dots,9$	weak	unsatisfiable	LRAT proof of CaDiCaL
identity	$2,\dots,12$	weak	unsatisfiable	DRAT proof of Kissat, turned into LRAT by drat-trim
identity	$2,\dots,11$	strong	satisfiable	the rule `balzer_sym11`

The conditions of Section 5 do not exclude every rule. For the germ of δ_{14}^{d}, which has complexity 15, conditions (G1)–(G3) with $T=10$ and all $n\ge 12$ have a model, and so do (G1)–(G3) with $T=7$ and all $n\ge 9$ together with (G5) for $9\le n\le 20$. In the second model, the lines of lengths 9, $11,\dots,17$ and 19 have both end cells, but not all cells, in state F at time $2n-2$, and no cell in state F before (Figure 18(a); both models are in the supplementary material). This suggests that excluding such rules needs conditions on the middle cells of the firing row, which are expensive to encode: for $n=15$, the dependency cone of the middle cell of the firing row contains 76 of the 120 cells of R_{15} (Figure 18(b)).

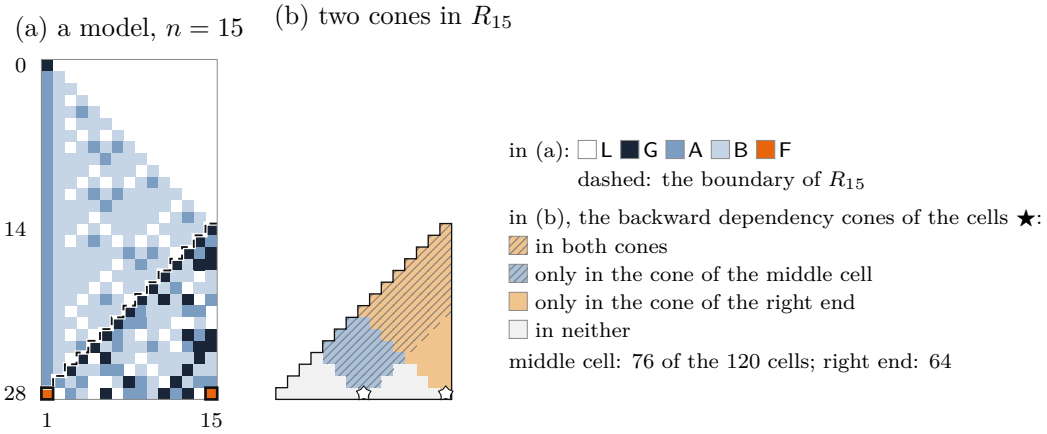


Figure 18: (a) The line of length 15 under the second model above, which contains the germ of δ_{14}^{d} and satisfies (G1)–(G3) with $T=7$ for all $n\ge 9$ and (G5) for $9\le n\le 20$. No cell is in state F before time $2n-2=28$, and at time 28 only the end cells 1 and 15 (outlined) are. (b) The backward dependency cones, inside R_{15}, of the middle cell ($\kappa=7$) and of the right end ($\kappa=0$) of the firing row (both marked ★), drawn level with (a). The first covers 76 of the 120 cells of R_{15}, the second 64; condition (G5) constrains only the second.

The computational part has three limitations. Its results concern four and five states and, apart from the reduction of Corollary 5.3, finitely many lengths. The formula generators are trusted programs, tested in Section 6.1 but not formally verified. And the behaviour of particular rules beyond the simulated ranges, such as the patterns of the half-lines of δ_{12} and δ_{14}, is not known.

We close with four questions.

(1) Does a five-state minimal-time solution exist?

(2) Is there a condition on the middle of the reflected triangles that is as economical as Lemma 5.5?

(3) Can the bound $(k-1)^{2s}\ell$ of Lemma 4.1 be made polynomial in s?

(4) Do Balzer's conditions exclude six states?

A THE ENCODINGS

Both formula generators build their formulas from the same pieces. Let S be the set of k states and $W = S \setminus \{\mathsf{F}\}$.

(E1) *Rule table.* A variable $x_{e,s}$ for each neighbourhood e and each $s \in S$, with the clause $\bigvee_s x_{e,s}$ and the clauses $\neg x_{e,s} \vee \neg x_{e,s'}$ for $s \neq s'$. The entries $\delta(\mathsf{L},\mathsf{L},\mathsf{L})$ and $\delta(\mathsf{L},\mathsf{L},*)$ are fixed to L.

(E2) *Cells.* A variable $y_{u,s}$ for each encoded cell u and each $s \in W$, with exactly-one clauses as in (E1). So every encoded cell is in a working state. The cells at time 0, the quiescent cells $t < i-1$ (Lemma 2.1) and the borders are constants, and so is every cell of a firing row, with the value F.

(E3) *Local rule.* For each encoded cell u with predecessors u_1, u_2, u_3 and each choice a, b, d of their values, let $e = (a, b, d)$ and let P be the conjunction of $y_{u_1,a}$, $y_{u_2,b}$ and $y_{u_3,d}$, constants omitted. If u is a working cell, the clauses are $P \wedge x_{e,s} \to y_{u,s}$ for $s \in W$ and $P \to \neg x_{e,\mathsf{F}}$; if u is in a firing row, the clause is $P \to x_{e,\mathsf{F}}$.

(E4) *Return chain.* For an encoded cell $u = (t, i)$ of the line of length n with $t + i = 2n - 1$ that is not in its firing row, and the half-line cell $h = (t, i)$: the clauses $\neg y_{u,s} \vee \neg y_{h,s}$ for $s \in W$.

(E5) *Pumping.* For PUMP(n, n'): the clause $\bigvee_p z_p$ over the positions p of $\mathrm{cone}_n(n-1, 0)$ (Lemma 3.2). The input cells v_p of J_n and w_p of $J_{n'}$ at p are compared by an auxiliary variable z_p with the clauses $\neg z_p \vee \neg y_{v_p,s} \vee \neg y_{w_p,s}$ for $s \in W$; constant cells are compared directly.

$\mathrm{MT}_k(\mathcal{N}; M, N')$ consists of (E1); the half-line cells with $t + i \le M$ and, for each $n \in \mathcal{N}$, the cells of R_n, with (E2) and (E3); (E4) for the lengths $n \in \mathcal{N}$ with $2n - 1 \le M$; and (E5) for $4 \le n < n' \le N'$. $\mathcal{G}_k(T, N_1, C)$ consists of (E1) and, with (E2)–(E4), the half-line cells with $t + i \le 2N_1$, the cells of R_n for $2 \le n \le T + 1$, the cells of R_n with $\tau \le T$ for $T + 2 \le n \le N_1$, and the cone of the right end at the firing time for $T + 2 \le n \le C$. These regions are closed under predecessors, so with the table fixed every encoded cell is determined. The clauses (E4) for the complete lines $2 \le n \le T + 1$ are not part of Definition 5.4, but they follow from (G1) and (G4) by the proof of Lemma 2.4, in which (G4) gives $C_\infty(2n - 2, 1) \neq \mathsf{F}$; so the formula is equivalent to Definition 5.4. For the formulas $\mathcal{G}_4$ of Theorem 6.1 and Proposition 6.2, the generator gives one variable to cells that compute the same function of the table, that is, cells with the same tree of predecessors down to the same half-line cells and borders; this does not change the satisfying assignments of the table variables. The formula of Proposition 6.4 adds, for each germ g, a selector s_g with the clauses $\neg s_g \vee x_{e,s}$ for the transitions (e, s) of g, and the clause $\bigvee_g s_g$. The conditions (B1)–(B4) of Proposition 6.6 are unit and binary clauses on the variables $x_{e,s}$.

B COMPUTATIONAL CLAIMS AND ARTEFACTS

Table 7 lists the computational claims of Section 6 with the programs behind them and the published files. The digests are SHA-256 digests of the uncompressed files, abbreviated to their first eight hexadecimal digits; the supplementary material lists them in full.

Table 7: The computational claims, their programs, and the unsatisfiable formulas with their LRAT proofs. The programs are in `src/` and `research/` of the supplementary material, and the formulas are the files `formulas/*.cnf.xz`. In the column Claim, (A B) etc. is the involution I of Proposition 6.6. The proof of the last formula was obtained from a DRAT proof of Kissat (digest `2dd3e052`, 958 MB) with drat-trim.

Claim	Programs	Formula	Variables	Clauses	SHA-256	LRAT proof
6.1 (i)	`gencnf.py`	`mt4_9`	753	17,605	`d18d6548`	`1d511a75`, 177 MB
6.1 (ii)	`gfix.py`	`k4_G6`	919	22,960	`0e0ba286`	`7943a698`, 167 MB
6.1 (iii)	`gfix.py`	`k4_G4c`	874	20,461	`124f0954`	`7f76fe69`, 173 MB
6.4	`germcert.py`	`k14_n10`	17,266	559,558	`4f9306a8`	`b2324436`, 11 MB
6.6, (A B)	`balzer.py`	`balzer_AB_10`	1,496	67,508	`d87457fe`	`18809db2`, 380 MB
6.6, (L A)	`balzer.py`	`balzer_LA_9`	1,244	49,571	`401ef851`	`d74de977`, 188 MB
6.6, (L B)	`balzer.py`	`balzer_LB_9`	1,244	49,571	`d7ba0d40`	`5784ee12`, 184 MB
6.6, id	`balzer_k.py`	`balzer_id_12`	2,148	114,965	`023b99e2`	`e0231f2f`, 3,078 MB
6.2 (a)	`fsspdfs.c` (enumeration)	—				
6.2 (b)	`gfix.py`; models checked by `encoding_check.py`	satisfiable formulas				
6.3	`fsspcheck.c`, `firstfire.py`, `failall.py`	simulations; certificates (t_0, p) in Table 5				
6.4	`germdfs.c`, `germdfs_check.py`, `germinc.py`, `germext.c`, `germ_k14_check.py`	enumeration of the 244 germs; the 14 refutations by Lemma 4.1				
6.5	`antidiag.py`, `germext.c`	simulation of the cells with $t + i \leq 1034$				
6.1	`encoding_check.py`, `validate_mazoyer.py`, `gfix.py`	checks of the encodings; Table 2				

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Use of generative AI The author used the AI system Claude (Anthropic) as an assistant. Under the author's direction, it assisted in checking and developing the arguments, in writing and running the programs and experiments, in drawing the figures, and in drafting and revising the text. The author reviewed and edited all content and takes full responsibility for it.

Data availability The programs, the rule tables, the germ lists and the logs behind every computational statement, the eight unsatisfiable formulas of Theorem 6.1 and Propositions 6.4 and 6.6 with the SHA-256 digests of the formulas and of their proofs, the three scripts that rerun every computation, the solver times and proof sizes, and the program that draws the figures are in the supplementary material and in the public repository `https://github.com/giamkiakeng/proj1`, directory `fssp5/paper/supplementary`. The version described in this paper is commit `1777ab2`, with the permanent link `https://github.com/giamkiakeng/proj1/tree/1777ab2a41df6bfacf2d56193660e7fe7906b460/fssp5/paper/supplementary`. The LRAT proofs, 4.4 GB in total, are too large for the repository; the third script regenerates them, checks them, and compares them with the recorded digests (Section 6.1 and Appendix B).

NOMENCLATURE

Symbol	Meaning	Where
$S,\ W,\ k$	the set of states, the working states $S \setminus \{\mathsf{F}\}$, and $k = \lvert S\rvert$	Section 2
$\mathsf{L},\ \mathsf{G},\ \mathsf{F}$	the quiescent state, the general and the firing state	Section 2
$\mathsf{A},\ \mathsf{B},\ \mathsf{C}$	auxiliary states	Section 2
$*$	the border	Section 2
δ	a rule, defined on the neighbourhoods (x, y, z)	Section 2
$C_n(t, i)$	the state of cell i at time t in the line of length n	Section 2
$C_\infty(t, i)$	the same in the half-line	Section 2
R_n	the reflected triangle of the line of length n	Section 2
$\tau,\ \kappa$	relative time $t-(n-1)$ and distance $n-i$ to the right end	Section 3
$R_n(\tau, \kappa)$	the state of C_n at $(n-1+\tau, n-\kappa)$	Section 3
$J_n(\kappa)$	the input word: the cells of the half-line on $t+i = 2n-3$ and $2n-2$	Section 3
Ψ	the map that computes R_n from J_n	Lemma 3.1
$\mathrm{cone}_n(\tau, \kappa)$	the input cells on which $R_n(\tau, \kappa)$ depends	Section 3
$\mathrm{PUMP}(n, n')$	the conclusion of the pumping lemma	Lemma 3.3
$\Phi_t(j)$	the cell of the half-line at depth j behind the front at time t	Section 3
$\theta(j)$	the time from which depth row j is periodic	Theorem 3.4
$\theta_T(j)$	the same, observed in a simulation up to time T	Remark 3.5
$v_c(i),\ u_c(y)$	the cells of chain c, in direct and in skewed coordinates	Section 4
$E(T)$	the first $T+1$ rows of a reflected triangle	Section 5
$V_t,\ \Gamma_D$	the first $D+1$ depth rows at time t, and the map that advances them	Lemma 5.2
$\mathcal{G}_k(T, N_1, C)$	the early-row conditions (G1)–(G5)	Definition 5.4
$\mathrm{MT}_k(\mathcal{N}; M, N')$	the formula for the lines in $\mathcal{N}$, the half-line up to $t+i = M$ and PUMP up to N'	Section 6.1
$c_M(\delta)$	the number of neighbourhoods besides $(\mathsf{L}, \mathsf{L}, \mathsf{L})$ on the half-line up to $t+i = M$	Section 6.4
$\delta_{12},\ \delta_{13},\ \delta_{14}$	five-state rules for the lengths up to 12, 13 and 14	Section 6.3

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